

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

TRANSGENE and BIOINVENT INTERNATIONAL AB,
Petitioner,

v.

REPLIMUNE LIMITED,
Patent Owner.

PGR2022-00014
Patent 10,947,513 B2

Before CHRISTOPHER G. PAULRAJ, RYAN H. FLAX, and
JAMIE T. WISZ, *Administrative Patent Judges*.

PAULRAJ, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 328(a)

I. INTRODUCTION

This is our Final Written Decision pursuant to 35 U.S.C. § 328(a) in the above-captioned proceeding. For the reasons explained herein, we determine that Transgene and BioInvent International AB (collectively, “Petitioner”) have not demonstrated by a preponderance of the evidence that challenged claim 13 of U.S. Patent 10,947,513 B2 (Ex. 1001, “the ’513 patent”) is unpatentable.

A. Background

On December 15, 2021, Petitioner filed a Petition requesting post-grant review of claims 1–8 and 10–26 of the ’513 patent. Paper 1 (“Pet.”). Replimune Limited (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 8 (“Prelim. Resp.”). As part of its Preliminary Response, Patent Owner included a Statutory Disclaimer of claims 1–8, 10–12, and 14–26 of the ’513 patent, leaving claim 13 as the sole remaining challenged claim in this proceeding. Prelim. Resp. 1; Ex. 2001. Pursuant to our authorization, Petitioner filed a Preliminary Reply addressing Patent Owner’s arguments for discretionary denial under 35 U.S.C. § 325(d). Paper 12 (“Prelim. Reply”). Patent Owner also filed a list of improper Preliminary Reply arguments. Paper 13.

In view of Patent Owner’s Statutory Disclaimer, we treat claims 1–8, 10–12, and 14–26 as having never been part of the ’513 patent, such that Petitioner cannot seek post-grant review of those claims. *See Vectra Fitness, Inc. v. TNWK Corp.*, 162 F.3d 1379, 1383 (Fed. Cir. 1998) (“This court has interpreted the term ‘considered as part of the original patent’ in section 253 to mean that the patent is treated as though the disclaimed claims never existed.”) (*citing Guinn v. Kopf*, 96 F.3d 1419, 1422 (Fed. Cir. 1996)); 37 C.F.R. § 42.207(e) (“No post-grant review will be instituted based on

disclaimed claims.”). On June 16, 2022, upon consideration of the information presented in the Petition, the Preliminary Response, the Preliminary Reply, and Patent Owner’s list of improper Preliminary Reply arguments, we instituted a post-grant review of the challenged claim. Paper 16 (“Inst. Dec.”).¹

Following institution, Patent Owner filed a Response to the Petition (Paper 22, “PO Resp.”), Petitioner filed a Reply to Patent Owner’s Response (Paper 24, “Reply”), and Patent Owner filed a Sur-Reply (Paper 26, “Sur-Reply”). On March 17, 2023, an oral argument was held in this proceeding, a transcript of which has been entered in the record. Paper 37 (“Tr.”). Following the oral argument, we authorized the parties to submit supplemental briefs on two recent decisions that bear on issues in this proceeding. Paper 34 (“Pet. Supp. Br.”); Paper 35 (“PO Supp. Br.”).

B. Real Parties-in-Interest

Petitioner identifies Transgene and BioInvent International AB as the real parties-in-interest (“RPI”). Pet. 1. Patent Owner identifies only itself as the RPI. Paper 6, 1. There is no dispute over RPIs in this proceeding.

C. Related Matters

The parties do not identify any related matters. *See* Pet. 2; Paper 6, 1.

¹ We based our decision to institute on the only remaining challenged claim (dependent claim 13), and thus did not substantively address the other claims challenged in the Petition. Inst. Dec. 3. Claim 13, however, depends upon claim 11, which in turn depends upon independent claim 1. As such, claim 13 incorporates the limitations of claims 1 and 11. 35 U.S.C. § 112(d) (“A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.”). As explained further below, notwithstanding the statutory disclaimer, the parties agree that it is necessary for us to first address the merits of the Petition’s challenge to claims 1 and 11 before turning to the merits of claim 13.

D. The '513 Patent (Ex. 1001)

The '513 patent is titled “Engineered Virus.” Ex. 1001, code (54). The '513 patent issued from application 16/068,830, filed January 9, 2017, and indicates foreign priority to three Great Britain applications filed January 8, 2016. *Id.* at codes (21), (22), (30).

The Specification explains that a number of viruses are known for the oncolytic treatment of cancer. *Id.* at 2:9–11. Viruses may treat cancer by directly infecting tumor cells, and by generating an immune response against tumor cells not infected with the virus. *See id.* at 1:22–46. Such a virus is known as an “oncolytic virus,” which is defined in the Specification as “a virus that infects and replicates in tumor cells, such that the tumor cells are killed.” *Id.* at 9:10–12. Notably, oncolytic viruses should be disabled so that they are no longer pathogenic. *See id.* at 2:11–14.

An oncolytic virus may be derived from various virus types, including herpes simplex virus (HSV). *See id.* at 2:9–11, 9:19–31. To allow for its use in the oncolytic treatment of cancer, “[a] number of disabling mutations to HSV, including disruption of the genes encoding [infected cell protein-34.5 (‘ICP34.5’)], . . . have been identified which do not prevent the virus from replicating in culture or in tumor tissue *in vivo*, but which prevent significant replication in normal tissue.” *Id.* at 2:14–19. The Specification further describes HSV oncolytic viruses that deliver a therapeutic gene for treating cancer. *Id.* at 2:25–27. For example, the Specification describes clinical trials that tested an HSV encoding a heterologous gene for granulocyte macrophage colony-stimulating factor (“GM-CSF”) in place of the gene for ICP34.5. *Id.* at 2:27–31. Furthermore, the Specification describes combining such oncolytic immunotherapy with immune checkpoint blockers, such as agents targeting anti-cytotoxic T lymphocyte-

associated antigen-4 (“CTLA-4”), programmed cell death protein-1 (“PD-1”), or programmed cell death protein ligand-1 (“PD-L1”). *See id.* at 2:49–3:11.

In view of this background, the ’513 patent discloses “oncolytic viruses expressing GM-CSF and at least one molecule targeting an immune co-stimulatory pathway.” *Id.* at 4:3–5. The immune co-stimulatory pathway activating molecule may include “a protein capable of blocking signaling through CTLA-4,” for example a CTLA-4 binding antibody. *Id.* at 4:65–67. “The virus is preferably [an HSV], such as HSV1” that “does not express functional ICP34.5.” *Id.* at 5:9–11.

The ’513 patent discloses examples of viruses expressing GM-CSF and CTLA-4. *See id.* at 30:43–31:14. Specifically, the Specification describes fusogenic viruses expressing murine GM-CSF (“mGM-CSF”), gibbon ape leukemia virus (“GALV”), and anti-mouse and anti-human CTLA-4. *Id.* The viruses expressing GM-CSF and CTLA-4 are Virus 27 and Virus 31, and their gene insertions are illustrated in Figures 5B, 5D, and 13. *Id.*

Figures 5B and 5D are reproduced, in part, below:

Figure 5B

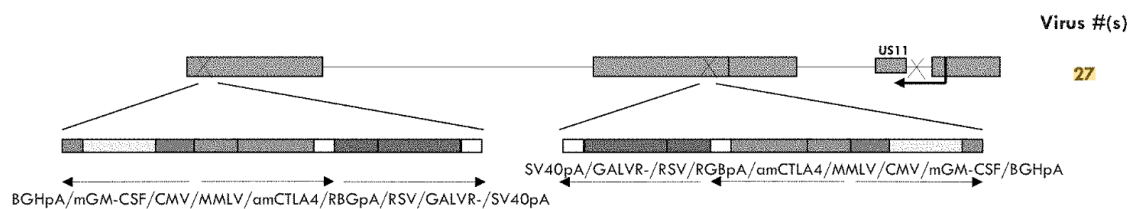
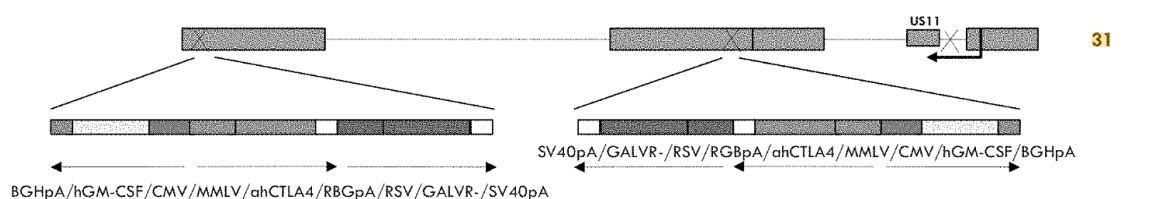


Figure 5D



Figures 5B and 5D “depict structures of HSV1 viruses modified by the deletion of ICP34.5 and ICP47 such that the US11 gene is under control of the ICP457 immediate early promoter and containing heterologous genes in the ICP34.5 locus.” *Id.* at 6:10–15. Figure 5B illustrates Virus 27. *See id.* at Fig. 5B. Figure 5D illustrates Virus 31. *See id.* at Fig. 5D.

Figure 13 is reproduced below:

Figure 13

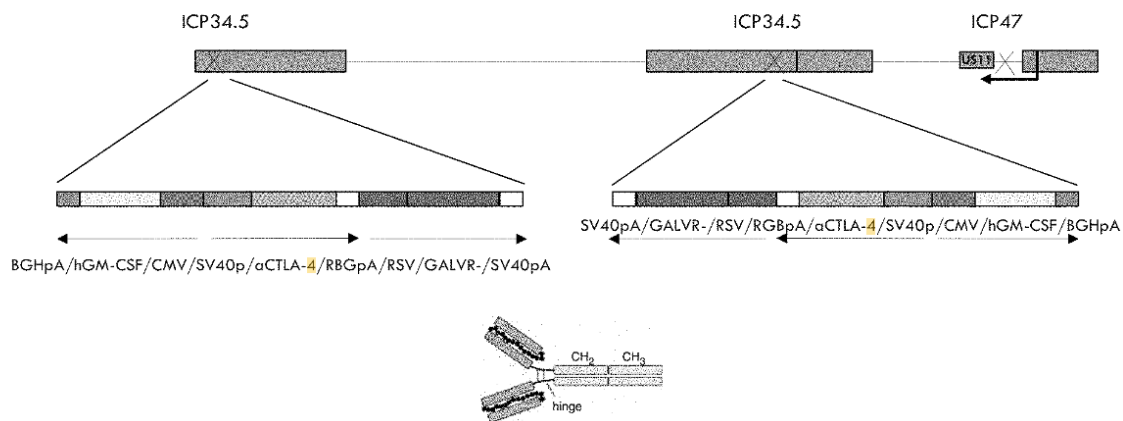


Figure 13 “shows the structure of ICP34.5 and ICP47 deleted viruses expressing . . . GM-CSF and codon optimized anti-mouse or anti-human CTLA-4 antibody constructs,” and the resulting structure of the CTLA-4 antibody. *Id.* at 7:9–17.

The '513 patent also discusses experiments comparing the effect of oncolytic viruses expressing anti-mouse CTLA-4 (Virus 27) or other immune co-stimulatory pathway activating molecules (Viruses 32, 33, and 35) in addition to mGM-CSF and GALV as compared to Virus 16 (which expressed only mGM-CSF and GALV). *Id.* at 30:45–31:12 (Example 13). According to the Specification, tumors were induced in both the right and left flanks of mice, but the oncolytic virus was injected only into the right

flank tumor. *Id.* at 31:5–12. The results are shown in Figure 16, reproduced below:

Figure 16

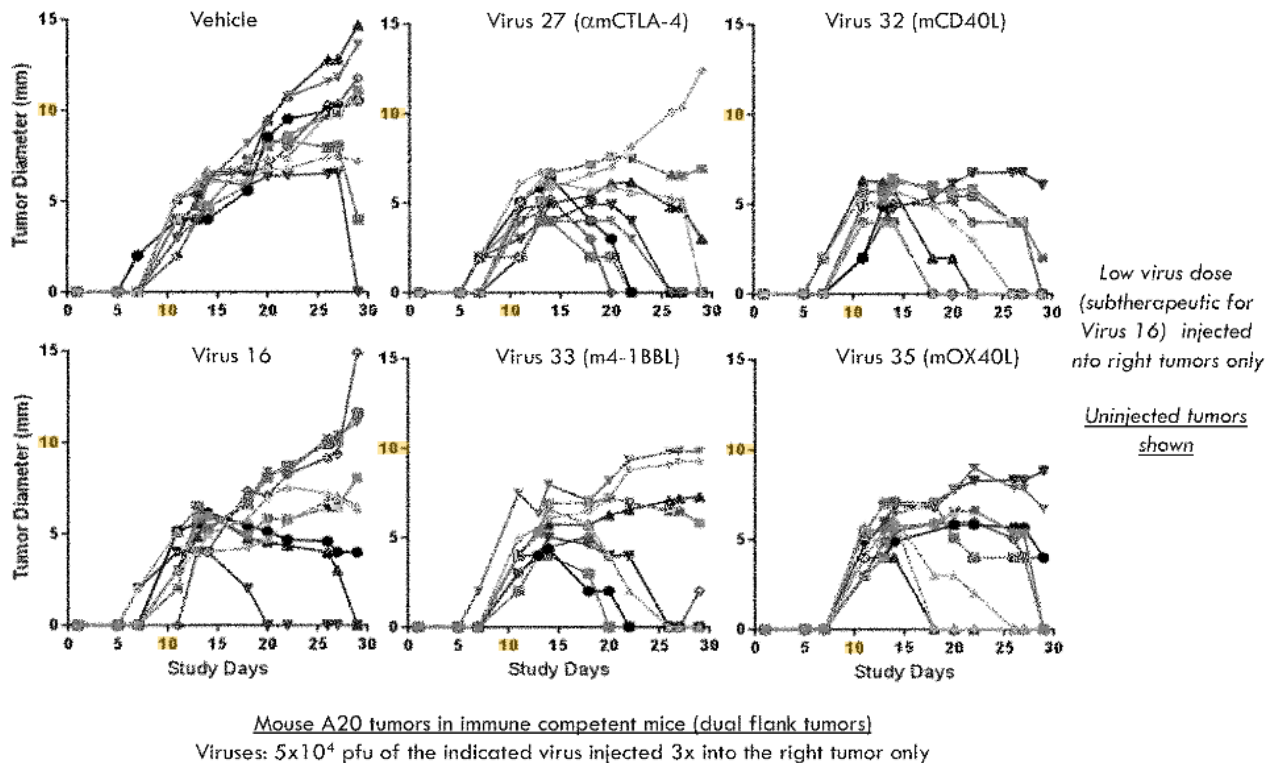


Figure 16 shows the anti-tumor effects of, *inter alia*, Virus 27 compared to Virus 16 on the tumor diameter of the uninjected left flank tumors over a period of 30 study days. *Id.* at 7:37–41.²

E. Illustrative Claims

Petitioner challenges claim 13 of the '513 patent, which depends on claim 11, which in turn depends on independent claim 1. We reproduce claims 1, 11, and 13 below:

1. An oncolytic virus comprising: (i) a heterologous GM-CSF-encoding gene; and (ii) a heterologous CTLA-4 inhibitor

² These results form the basis for Patent Owner's unexpected results arguments, and are further discussed in our analysis below.

encoding gene, wherein both heterologous genes are inserted into the genome of the virus.

11. The virus of claim 1, which is a herpes simplex virus (HSV).

13. The virus of claim 11, wherein the GM-CSF-encoding gene and the CTLA-4 inhibitor encoding gene are inserted into the ICP34.5 encoding locus, either by insertion, or partial or complete deletion, in a back to back orientation in relation to each other, each under separate regulatory control.

Ex. 1001, 81:51–54; 83:17–18, 20–25.

F. Prior Art and Asserted Grounds

Petitioner asserts that claim 13 would have been unpatentable on the following grounds:

Claim(s) Challenged	35 U.S.C. §³	Reference(s)/Basis
1, 11 ⁴	102	Silvestre ⁵
13	103	Silvestre

³ The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284 (2011), amended 35 U.S.C. § 102 and § 103, effective March 16, 2013. Because the effective filing date of the challenged claims of the ’513 patent is after this date, the AIA versions of § 102 and § 103 apply.

⁴ Although claims 1 and 11 are disclaimed, we include and address the grounds upon which those claims were challenged in the Petition before turning to claim 13 because claim 13 includes the subject matter of claims 1 and 11. *See Intel Corp. v. PACTXPP Schweiz AG*, 61 F.4th 1373, 1375 (Fed. Cir. 2023) (“[T]o show that [the challenged dependent claim] was unpatentable, [petitioner] still had to demonstrate that the prior art taught all limitations in [the disclaimed independent claim] (in addition to the limitations added by [the dependent claim]) because [the dependent claim] includes all of the limitations of that underlying independent claim”); Tr. 17:20–18:13 (Petitioner’s counsel acknowledging that Petitioner still had the burden to show that claims 1 and 11 were anticipated or rendered obvious based on the grounds set forth in the Petition for those claims).

⁵ Silvestre et al., US 10,555,981 B2, issued Feb. 11, 2020 (Ex. 1002).

Claim(s) Challenged	35 U.S.C. § ³	Reference(s)/Basis
1, 11, 13	103	Du ⁶ , Choi ⁷ , Zitvogel ⁸

Petitioner relies on the declaration of John C. Bell, Ph.D. (Ex. 1007).
Patent Owner relies on the declaration of E. Antonio Chiocca, M.D., Ph.D. (Ex. 2009).

II. DISCUSSION

A. Legal Standards

1. Anticipation

In order to anticipate a combination invention, “disclosure of each element is not quite enough;” rather, “[a]nticipation requires the presence in a single prior art disclosure of all elements of a claimed invention arranged as in the claim.” *Finisar Corp. v. DirecTV Grp., Inc.*, 523 F.3d 1323, 1334 (Fed. Cir. 2008). That is, the prior art reference must “clearly and unequivocally disclose the claimed [invention] or direct [POSAs] to [it] without *any* need for picking, choosing, and combining various disclosures not directly related to each other by the teachings of the cited reference.” *In re Arkley*, 455 F.2d 586, 587 (CCPA 1972). “[A] reference can anticipate a claim even if it ‘does not expressly spell out’ all the limitations arranged or combined as in the claim, if a person of ordinary skill in the art, reading the reference, would ‘at once envisage’ the claimed arrangement or combination.” *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d

⁶ T. Du et al., *Tumor-specific oncolytic adenoviruses expressing granulocyte macrophage colony-stimulating factor or anti-CTLA4 antibody for the treatment of cancers*, 21 CANCER GENE THERAPY 340–48 (2014) (Ex. 1003).

⁷ K-J Choi et al., *Concurrent delivery of GM-CSF and B7-1 using an oncolytic adenovirus elicits potent antitumor effect*, 13 GENE THERAPY 1010–20 (2006) (Ex. 1005).

⁸ Zitvogel et al., US 10,765,710 B2, issued Sept. 8, 2020 (Ex. 1004).

1376, 1381 (Fed. Cir. 2015) (“*Kennametal*”) (quoting *In re Petering*, 301 F.2d 676, 681 (CCPA 1962). “The question for purposes of anticipation is therefore whether the number of categories and components [in the reference is] so large that the [claimed] combination . . . would not be immediately apparent to one of ordinary skill in the art.” *Wm. Wrigley Jr. Co. v. Cadbury Adams USA LLC*, 683 F.3d 1356, 1361 (Fed. Cir. 2012).

2. Obviousness

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) where in evidence, so-called objective indicia or secondary considerations, including commercial success, long-felt but unsolved needs, failure of others, and unexpected results (“the *Graham* factors”). *Graham v. John Deere Co. of Kansas City*, 383 U.S. 1, 17–18 (1966). When modifying prior art teachings, “there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418 (quoting *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)).

“For objective evidence of secondary considerations to be accorded substantial weight, its proponent must establish a nexus between the evidence and the merits of the claimed invention.” *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1246 (Fed. Cir. 2010)). “[A] patentee is entitled to a rebuttable presumption of nexus between the asserted evidence of secondary

considerations and a patent claim if the patentee shows that the asserted evidence is tied to a specific product and that the product ‘is the invention disclosed and claimed.’” *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (quoting *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1392 (Fed. Cir. 1988)). A presumption of nexus based on a commercial product is not appropriate when the product “is not coextensive with the patented invention,” but such a presumption may be appropriate if the product includes “unclaimed features [that] amount to nothing more than additional insignificant features.” *Id.* at 1373–74 (quoting *Demaco*, 851 F.2d at 1392). “A finding that a presumption of nexus is inappropriate does not end the inquiry into secondary considerations”; rather, “the patent owner is still afforded an opportunity to prove nexus by showing that the evidence of secondary considerations is the ‘direct result of the unique characteristics of the claimed invention.’” *Id.* at 1374–75 (quoting *In re Huang*, 100 F.3d 135, 140 (Fed. Cir. 1996)).

B. Level of Ordinary Skill in the Art

Petitioner proposes that a person of ordinary skill in the art (“POSA” or “POSITA”) “would have had a Ph.D. in molecular biology, or a related field, with expertise in virology (including expertise growing, isolating, and rescuing viruses), immunology, and cancer biology with at least four years of post-Ph.D. experience in those areas.” Pet. 6 (citing Ex. 1007 ¶¶ 35, 96). Patent Owner does not argue otherwise. *See* PO Resp. 5.

We adopt Petitioner’s uncontested definition as that definition is consistent with the level of skill in the art reflected by the prior art. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not

shown” (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985))). It is from the perspective of such a person of ordinary skill that we consider the claims and prior art.

C. Claim Construction

We interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.200(b) (2019). This standard requires that we construe claims “in accordance with the ordinary and customary meaning of such claim[s] as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

Petitioner contends that the Specification defines the term “‘oncolytic virus’ as ‘a virus that infects and replicates in tumor cells, such that the tumor cells are killed.’” Pet. 6 (citing Ex. 1001, 9:10–12). Petitioner argues that the Specification “distinguishes this broader definition from the allegedly narrower subcategory of ‘selectively replication competent’ viruses that ‘replicate[] more effectively in tumor tissue than in non-tumor tissue.’” *Id.* (citing Ex. 1001, 9:12–17).

In our Institution Decision, we did not perceive a need to construe any claim terms of the ’513 patent for purposes of determining whether to institute trial. For purposes of our analysis in this Final Written Decision, however, we find that it would be useful to clarify the scope of “oncolytic virus.” We agree with Petitioner that the Specification defines an “oncolytic virus” as “a virus that infects and replicates in tumor cells, such that tumor cells are killed.” Ex. 1001, 9:10–12. The Specification further clarifies that “the virus is replication competent,” and is preferably selectively replication competent in tumor tissue. *Id.* at 9:12–13. “A patentee is free to be his own lexicographer.” *Mentor Graphics Corp. v. EVE-USA, Inc.*, 851 F.3d 1275,

1294 (Fed. Cir. 2017). Accordingly, we construe “oncolytic virus,” consistent with its express definition in the Specification, to be a virus that infects and replicates in tumor cells, such that tumor cells are killed, wherein such a virus is replication competent.

Patent Owner does not dispute this aspect of Petitioner’s construction, but contends that “oncolytic virus” should not be construed to encompass “viral vectors” such as those disclosed in Zitvogel. PO Resp. 5. Although Petitioner does not expressly contend otherwise, we agree with Patent Owner that “oncolytic virus” would not encompass Zitvogel’s “viral vectors,” which are defined in that reference to be “replication-defective” or “replication-impaired.” Ex. 1004, 19:26–29.

We do not construe any other claim terms. *See Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 295, 803 (Fed. Cir. 1999) (holding that “only those terms need [to] be construed that are in controversy, and only to the extent necessary to resolve the controversy”); *see also Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (applying *Vivid Techs.* in the context of an *inter partes* review).

D. Overview of Prior Art

1. Silvestre (Ex. 1002)

Silvestre is a U.S. patent that issued on February 11, 2020, from Application No. 15/325,562, that was filed as PCT/EP2015/066263 on July 16, 2015. Ex. 1002, codes (21), (22), (45), (86). Thus, on its face, Silvestre qualifies as prior art under 35 U.S.C. § 102(a)(2). Patent Owner does not dispute Silvestre’s status as prior art.⁹

⁹ Silvestre is assigned to Petitioner Transgene SA, and further lists Transgene SA as the applicant for the patent that issued as Silvestre. Ex. 1002, code (73).

Silvestre is directed to an oncolytic virus including a nucleotide sequence that encodes an immune checkpoint modulator. *Id.* at 3:23–26. The oncolytic virus is preferably selected from a group that includes HSV. *Id.* at 3:27–31. The immune checkpoint modulator preferably includes an antagonist molecule that antagonizes the activity of CTLA-4, “with a specific preference for an anti PD-1 antibody and/or an anti CTLA4 antibody.” *Id.* at 3:41–48; *see also id.* at 15:5–12, 45–48.

Silvestre discloses further embodiments of oncolytic viruses. For example, the oncolytic virus may include a herpes virus derived from HSV1. *Id.* at 7:52–60. “The herpes virus may be genetically modified so as to restrict viral replication in tumors or reduce its cytotoxicity in non-dividing cells.” *Id.* at 7:60–63. For example, the virus may include “defects in the function of genes encoding virulence factors such as the ICP34.5 gene.” *Id.* at 8:1–3.

Silvestre discloses oncolytic viruses including immunostimulatory proteins. *Id.* at 11:55–57. “A vast number of proteins are known in the art for their ability to exert an immunostimulatory effect.” *Id.* at 11:57–59. Silvestre describes a “specific preference for GM-CSF” as an immunostimulatory protein. *Id.* at 12:7–9. Silvestre further describes a known “oncolytic herpes simplex 1 (T-VEC) . . . genetically engineered to attenuate the virus virulence, increase selectivity for cancer cells and enhance antitumor immune response (through GM-CSF expression).” *Id.* at 2:32–30.¹⁰

¹⁰ Citing Neil N. Senzer et al., *Phase II Clinical Trial of a Granulocyte-Macrophage Colony-Stimulating Factor–Encoding, Second-Generation Oncolytic Herpesvirus in Patients With Unresectable Metastatic Melanoma*, 27(34) J. CLIN. ONCOLOGY 5763–71 (2009) (Ex. 1078).

Silvestre discloses that “[t]he nucleic acid molecule(s) encoding the immune checkpoint modulator(s) and eventually the therapeutic gene(s) can independently be inserted at any location of the viral genome, with a specific preference for a non-essential locus.” *Id.* at 16:4–8.

2. *Du (Ex. 1003)*

Du is a journal article that bears a publication date of 2014. Ex. 1003, 340. Petitioner contends that Du was published and accessible to the public no later than July 18, 2014. Pet. 5 (citing Ex. 1011 ¶¶ 47–55). Thus, on its face, Du qualifies as prior art under 35 U.S.C. § 102(a)(1). *Id.* Patent Owner does not dispute Du’s status as prior art.

Du discloses oncolytic adenoviruses expressing either GM-CSF or anti-CTLA4 antibody for the treatment of cancers. Ex. 1003, 340. Du discloses two viruses in particular, SKL001 and SKL002. *Id.* at 341. SKL001 encodes the heavy and light chains of an anti-CTLA-4 monoclonal antibody (“anti-CTLA-4”). *Id.* SKL002 replaces the coding region of anti-CTLA-4 with the cDNA of GM-CSF. *Id.* at 341. Du discloses that administering both viruses in combination, was “significantly more effective” in treating tumors than administering each virus alone. *Id.* at 345–346.

3. *Choi (Ex. 1005)*

Choi is a journal article that bears a publication date of 2006. Ex. 1005, 1010. Petitioner contends that Choi was published and accessible to the public no later than March 9, 2006. Pet. 5 (citing Ex. 1011 ¶¶ 38–46). Thus, on its face, Choi qualifies as prior art under 35 U.S.C. § 102(a)(1). *Id.* Patent Owner does not dispute Choi’s status as prior art.

Choi discloses an oncolytic adenovirus that expresses both GM-CSF and B7-1, an immune-stimulatory membrane molecule. *Id.* at 1010. Choi

discloses preparing the adenovirus by placing two independent expression cassettes, each separately expressing murine GM-CSF and murine B7-1, in the E1 and E3 region of a replication-incompetent adenovirus. *Id.* at 1011.

4. *Zitvogel (Ex. 1004)*

Zitvogel is a U.S. patent that issued on September 8, 2020 from Application No. 15/325,576, that was filed as PCT/EP2015/066353 on July 16, 2015. *Ex. 1004*, codes (21), (22), (45), (86). Thus, on its face, *Zitvogel* qualifies as prior art under 35 U.S.C. § 102(a)(2). Patent Owner does not dispute *Zitvogel*'s status as prior art.

Zitvogel discloses cancer therapy including a combination of an oncolytic virus with an immune checkpoint modulator. *Id.* at 3:30–33. The oncolytic virus is preferably selected from the group including vaccinia virus and HSV. *Id.* at 3:33–38. *Zitvogel* discloses an embodiment in which the oncolytic virus is an HSV mutant with defects in the ICP34.5 gene, including T-VEC as a representative example. *See id.* at 8:10–35.

Zitvogel discloses that the immune checkpoint modulator may preferably be an anti-CTLA-4 antibody. *Id.* at 3:51–53. *Zitvogel* discloses that the immune checkpoint modulator “may be delivered to the subject in the form of a vector expressing the one or more immune checkpoint modulator.” *Id.* at 22:37–43.

Zitvogel discloses an embodiment where the oncolytic virus expresses at least one therapeutic gene encoding an immunostimulatory protein. *Id.* at 3:48–50. “Preferably, the [immunostimulatory] protein is an interleukin or a colony-30 stimulating factor, with a specific preference for GM-CSF.” *Id.* at 12:28–30. *Zitvogel* discloses that the therapeutic gene is preferably inserted in a non-essential locus of the viral genome. *Id.* at 13:60–63.

E. Grounds Based on Silvestre

1. Analysis of Claims 1 and 11

In its Petition, Petitioner asserts that claims 1 and 11 are anticipated by Silvestre because “Silvestre discloses each element of these claims and Silvestre’s disclosure is enabling and describes the claimed invention sufficiently to have placed it in possession of a POSITA.” Pet. 42.

Petitioner then identifies where each element of independent claim 1 is taught by Silvestre. *Id.* at 42–43. Specifically, with regard to independent claim 1, Petitioner contends that Silvestre teaches an oncolytic virus (Ex. 1002, 2:23–26, 3:31–32) comprising (i) a heterologous GM-CSF encoding gene (*id.* at 10:8–10, 10:28–31, 12:7–9), and (ii) a heterologous CTLA-4 inhibitor encoding gene (*id.* at 3:44–48, 15:5–8), wherein both heterologous genes are inserted into the genome of the virus (*id.* at 3:23–26, 16:4–8). *Id.* at 42–43. With regard to dependent claim 11, Petitioner contends that Silvestre teaches that the oncolytic virus may be an HSV (Ex. 1002, 7:52–60). Pet. 48.

As to these anticipation arguments, Patent Owner contends that Silvestre does not disclose the limitations as arranged in claims 1 and 11. PO Resp. 11–17. In particular, Patent Owner contends that Silvestre focuses on utilizing a vaccinia virus (i.e., a poxvirus), rather than HSV, as the oncolytic virus, and that the claims and examples require that the immune checkpoint modulator is an antibody that binds to PD-1, rather than CTLA-4. *Id.* at 12–14. More generically, Patent Owner contends that Silvestre lists 11 possible oncolytic viruses, 14 immune checkpoint modulators, and 33 therapeutic genes, resulting in over 5,000 different possible combinations. *Id.* at 15 (citing Ex. 1002, 7:10–15:49; Ex. 2009 ¶ 29). Patent Owner contends that Petitioner provides “no reasoned analysis or explanation as to

how, based on the disclosure of Silvestre, a POSA would **at once envisage** the specific oncolytic virus claimed in claims 1 and 11.” *Id.* at 16.

Petitioner, in its Reply, contends that Silvestre emphasizes only twelve options, not “over 5,000.” Reply 2–7. To arrive at this more limited number of viruses, Petitioner points out that Silvestre identifies only two of the eleven listed viruses as “preferred,” with HSV1 and HSV2 identified as “particularly preferred.” *Id.* at 3–4 (citing Ex. 1002, 7:52–60). Petitioner also contends that a POSITA would have immediately envisaged only three viruses (herpesvirus, poxvirus, and adenovirus) as suitable for the insertion of multiple heterologous genes. *Id.* at 4. Petitioner further notes that Silvestre identifies only two preferred immune checkpoint modulators (anti PD-1 antibody and/or anti-CTLA-4 antibody), and also identifies only two preferred therapeutic genes (with a specific preference for GM-CSF). *Id.* at 5–6. Petitioner contends that “the twelve combinations from Silvestre immediately envisaged by a POSITA, coupled with the specific preferences for each of HSV, GM-CSF, and anti CTLA-4 antibody, align with the anticipation finding in” *Kennametal*, 780 F.3d 1376. *Id.* at 8 (also citing *In re Petering*, 301 F.2d 676 (CCPA 1962); *In re Schaumann*, 572 F.2d 312, 314 (CCPA 1978)).

Patent Owner, in its Sur-Reply, contends that Petitioner’s reliance on this *Kennametal*-type argument in its Reply is improper and should be disregarded. PO Sur-Reply 1–2. But even if considered, Patent Owner contends that Petitioner’s reliance on *Kennametal* is misplaced insofar as the Board “had expressly rejected the argument that options characterized as ‘preferred’ somehow negated the description of others.” *Id.* at 4. Patent Owner contends that Petitioner’s narrowed list of viruses is based on impermissible hindsight, and that Silvestre does not disclose that HSV is one

of only two virus types preferred by Silvestre. *Id.* at 5–6. Patent Owner argues that Petitioner’s new rationale for narrowing Silvestre’s list of viruses is based on “packaging capacity,” which is not a concept relied upon in the Petition in support of anticipation, and in any event, there is no basis to limit Silvestre’s list of viruses based on such common “knowledge in the art.” *Id.* at 7–9. Patent Owner also contends that Silvestre identifies more than two preferred therapeutic genes and more than two preferred immune checkpoint modulators. *Id.* at 9–14.

Having considered all the arguments and evidence of record, we determine that Petitioner has not established by a preponderance of the evidence that Silvestre anticipates claim 11.¹¹

With regard to claims specifically reciting HSV (including claim 11), Petitioner acknowledges that “while the bulk of Silvestre’s disclosure, including the working examples, is directed towards creating an oncolytic poxvirus, Silvestre discloses preparing modified oncolytic virus using other virus types, including HSV.” Pet. 44. It is undisputed, however, that Silvestre does not disclose a single embodiment of an oncolytic HSV that includes both a heterologous GM-CSF encoding gene and a heterologous CTLA-4 inhibitor encoding gene. Rather, Petitioner points to where each element could be found among broader lists of possible oncolytic virus classes, immune checkpoint targets, and therapeutic genes. Ex. 1002, 7:10–15:49. As noted by Dr. Chiocca, if one were to randomly select any of the

¹¹ We need not decide whether Silvestre anticipates independent claim 1, as the sole challenged claim 13 depends directly from claim 11. Thus, our determination that Petitioner has not shown that claim 11 is anticipated is sufficient to establish that Petitioner has not met its prerequisite burden for claim 13.

disclosed options, this could result in thousands of possible combinations. Ex. 2009 ¶ 29. This type of picking and choosing is not anticipation. *See Arkley*, 455 F.2d at 587–88.

In its Reply, Petitioner contends that Silvestre nonetheless anticipates because a POSITA would have immediately envisaged the claimed combination from among the “small, recognizable class of twelve options” taught by Silvestre. Reply 3–8. Petitioner never identified this smaller set of twelve options in the Petition, and it is not appropriate to raise new unpatentability theories for the first time in the Reply. *See* 37 C.F.R. § 42.23(b) (“A reply may only respond to arguments raised in the corresponding opposition, patent owner preliminary response, or patent owner response.”) (emphasis added); *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1369–70 (Fed. Cir. 2016) (holding that the Board did not err in refusing the reply brief as improper under 37 C.F.R. § 42.23(b) because petitioner relied on an entirely new rationale to explain why one of skill in the art would have combined the references at issue). We therefore agree with Patent Owner that Petitioner’s reliance on *Kennametal* (and related cases) to belatedly contend that a POSITA would have immediately envisaged the claimed combination from among a smaller set of options within Silvestre’s broad disclosures is a new unpatentability theory.¹² *See Wasica Fin. GmbH v. Cont’l Auto. Sys.*, 853 F.3d 1272, 1285-

¹² Although Patent Owner did not file a motion to strike Petitioner’s Reply on this basis, our Trial Practice Guide states that “the Board is capable of identifying new issues or belatedly presented evidence when weighing the evidence at the close of trial, and disregarding any new issues or belatedly presented evidence that exceeds the proper scope of reply or sur-reply” regardless of whether a motion to strike is filed. *See* Patent Trial and Appeal Board Consolidated Trial Practice Guide, 80 (Nov. 2019) (“CTPG”),

86 (Fed. Cir. 2017) (“To be sure, we have recognized instances where a prior art genus may anticipate a later species, *see, e.g., Kennametal* But in the ‘295 proceeding, Continental’s petition failed to set forth the necessary ‘factual component[s]’ needed to advance this legal theory.”); *Broad Ocean v. Nidec Motor*, IPR2015-01617, Paper 70 (Second Final Written Decision) at 23 (PTAB April 25, 2019) (“Petitioner’s reliance on the reasoning of *Kennametal* is improper because it was not raised in the Petition.”).

Even if we considered Petitioner’s new theory, we do not find it sufficient. Specifically, to arrive at the narrowed list of viruses, Petitioner contends that Silvestre teaches that HSV1 and HSV2 are “particularly preferred.” Reply 4. But the full context in which that statement is made in Silvestre is as follows:

In one embodiment, the oncolytic virus of the present invention is obtained from a herpes virus. . . . Although the oncolytic herpes virus can be derived from different types of HSV, particularly preferred are HSV1 and HSV2.

Ex. 1002, 7:52–60. Elsewhere, Silvestre identifies up to eleven additional virus types as either “preferabl[e]” or “exemplary”:

The oncolytic virus is preferably selected from the group consisting of reovirus, New Castle Disease virus (NDV), vesicular stomatitis virus (VSV), measles virus, influenza virus, Sinbis virus, adenovirus, poxvirus and herpes virus (HSV) and the like.

Id. at 3:27–31;

Exemplary oncolytic viruses include without limitation reovirus, Seneca Valley virus (SVV), vesicular stomatitis virus (VSV), Newcastle disease virus (NDV), herpes simplex virus

available at <https://www.uspto.gov/TrialPracticeGuideConsolidated>. We do so here.

(HSV), morbillivirus virus, retrovirus, influenza virus, Sin bis virus, poxvirus, adenovirus, or the like.

Id. at 7:26–30.

Contrary to Petitioner’s arguments, we find nothing in Silvestre to suggest that HSV is *more* preferable than any of the other listed virus types. Rather, based on the entirety of Silvestre’s teachings, we find that a POSITA would have understood Silvestre’s statement that “particularly preferred are HSV1 and HSV2” to mean that those two particular strains of HSV are only preferred if the HSV is selected as the virus type.¹³ That does not suggest, however, that HSV—as a general class of viruses—is more preferred than the other classes of viruses listed in Silvestre. To the extent Silvestre exhibits a higher preference for any particular virus type among the general categories listed, it is for poxviruses (and in particular the vaccinia virus). *See, e.g., id.* at 8:43–44 (“Desirably, the oncolytic poxvirus is an oncolytic vaccinia virus.”); *id.* at 26:60–67 (examples utilizing poxvirus); *id.* at 37:33–34 (claims requiring vaccinia virus). As explained by Dr. Chiocca, “when [Silvestre] talks about the ‘particularly preferred,’ what they’re saying is that because there’s multiple strains of HSV, the particularly preferred ones are HSV-1 and 2. It doesn’t say that HSV is a preferred virus.” Ex. 1106, 116:12–20; *see also id.* at 117:11–17 (“[A]gain, herpes virus is one embodiment just like everybody else, amongst that embodiment, HSV-1 and 2 are preferred. It’s not that they’re preferred, they’re preferred within that embodiment.”). Petitioner does not present any expert testimony that contradicts Dr. Chiocca’s interpretation of Silvestre, but instead relies upon unsupported attorney argument. *See Icon Health & Fitness, Inc. v. Strava*,

¹³ As noted by Petitioner, there are 941 known strains of HSV. Pet. 26 (citing Ex. 1023, Ex. 1024).

Inc., 849 F.3d 1034, 1043 (Fed. Cir. 2017) (“Attorney argument is not evidence.”).

We are also not persuaded by Petitioner’s contention that a POSITA would have focused on only “three realistic virus options” from among Silvestre’s list of eleven viruses for the insertion of multiple heterologous genes back in 2016, i.e., herpesvirus, poxvirus, and adenovirus. Reply 4. We agree with Patent Owner that “there is simply no reason for the POSA to limit Silvestre’s list of viruses to only those that were purportedly commonly used to express multiple heterologous genes, as Silvestre discloses embodiments where the virus encodes a single heterologous gene.” Sur-Reply 5; Ex. 1002, 3:23–26 (“The present invention concerns an oncolytic virus comprising inserted in its genome one or more nucleic acid molecule(s) encoding one or more immune checkpoint modulator(s).”). Petitioner cites testimony of Dr. Bell in support, but Dr. Bell does not address the teachings of Silvestre in the cited portions of his declaration; nor does not Dr. Bell assert that a POSITA would have only considered herpesvirus, poxvirus, and adenovirus among the viruses taught by Silvestre for the insertion of multiple heterologous genes. Reply 4 (citing Ex. 1007 ¶¶ 61–72, 104–107). To the extent Petitioner focuses on only those viruses that would allow for insertion of multiple genes, it is based on improper picking and choosing. *See In re Ruschig*, 343 F.2d 965, 974 (CCPA 1965) (“We did not intend our *Petering* opinion or decision to become a precedent for the mechanistic dissection and recombination of the components of the specific illustrative compounds in every chemical reference containing them, to create hindsight anticipations with the guidance of an applicant’s disclosures, on the theory that such reconstructed disclosures describe specific compounds within the meaning of section 102.”).

We have also considered the testimony of Dr. Chiocca cited by Petitioner, but do not find that he admitted that a POSITA would have envisaged only herpesvirus, poxvirus, and adenovirus to the exclusion of other viruses disclosed by Silvestre. For example, Dr. Chiocca answered “[c]orrect” when asked if “a person of ordinary skill in the art looking at that list in 2016 would have been able to immediately envisage that you could use herpes[virus], pox[virus], or adeno[virus] from this list.” Ex. 1106, 111:17–21. But Dr. Chiocca did not state that those were the only three viruses that a POSITA would immediately envisage for the insertion of multiple heterologous genes, and instead expressed some uncertainty as to whether other viruses could also be utilized in the same manner. *Id.* at 111:2–9 (indicating that he would have to look up whether measles, retrovirus, and influenza would also be used for insertion of heterologous genes). Moreover, even assuming a POSITA would have focused on only three of the eleven categories of viruses, Petitioner has not demonstrated that a POSITA would have at once envisaged the claimed oncolytic virus among a narrow set in view of the total combination of variables disclosed in Silvestre.

Likewise, we are not persuaded that a POSITA would have only focused on GM-CSF among the therapeutic genes taught by Silvestre. Although Petitioner quotes Silvestre’s statement indicating a “specific preference for GM-CSF,” the full context in which that statement appears is as follows:

Examples of *suitable immunostimulatory proteins* in the context of the invention include without limitation cytokines, with a *specific preference for interleukins* (e.g. IL-2, IL-6, IL-12, IL-15, IL-24), chemokines (e.g. CXCL10, CXCL9, CXCL11), interferons (e.g. IFN γ , IFN α), tumor necrosis factor (TNF),

colony-stimulating factors (e.g. GM-CSF, C-CSF, M-CSF ...), APC (for Antigen Presenting Cell)-exposed proteins (e.g. B7.1, B7.2 and the like), growth factors (Transforming Growth Factor TGF, Fibroblast Growth Factor FGF, Vascular Endothelial Growth Factors VEGF, and the like), MHC antigens of class I or II, apoptosis inducers or inhibitors (e.g. Bax, Bcl2, BclX ...), cytostatic agents (p21, p16, 5 Rb ...), immunotoxins, antigenic polypeptides (antigenic polypeptides, epitopes, and the like) and markers (betagalactosidase, luciferase ...). *Preferably, the immunostimulatory protein is an interleukin or a colony-stimulating factor*, with a specific preference for GM-CSF.

Ex. 1002, 11:53–12:9 (emphasis added). As a matter of syntax, the reference to a “specific preference for GM-CSF” in the last sentence of the foregoing paragraph relates back to “colony-stimulating factor,” meaning that GM-CSF is a preferred “colony-stimulating factor” over other types disclosed (such as C-CSF and M-CSF), not that GM-CSF is Silvestre’s only preferred immunostimulatory protein. As explained by Dr. Chiocca, “there’s lots of preferreds in that sentence.” Ex. 1106, 129:2–9. Petitioner does not explain why a POSITA “would have focused” only on GM-CSF while ignoring all of the other “preferred” immunostimulatory proteins disclosed by Silvestre.

At best, Petitioner demonstrates that a POSITA would have found it obvious to pick and choose amongst the various components taught by Silvestre to arrive at the claimed invention, but this is insufficient to establish anticipation. *See Arkley*, 455 F.2d at 587–88 (“Such picking and choosing may be entirely proper in the making of a 103, obviousness rejection, where the applicant must be afforded an opportunity to rebut with objective evidence any inference of obviousness which may arise from the similarity of the subject matter which he claims to the prior art, but it has no

place in the making of a 102, anticipation rejection.”). Petitioner has not made an obviousness argument for claims 1 and 11 based on Silvestre.¹⁴

2. *Analysis of Claim 13*

With respect to dependent claim 13, Petitioner does assert obviousness as part of Ground 4, and provides the following rationale in the Petition (reproduced in its entirety):

Claim 13 depends from Claim 11 and requires that “the GM-CSF-encoding gene and the CTLA-4 inhibitor encoding gene are inserted into the ICP34.5 encoding locus, either by insertion, or partial or complete deletion, in a back to back orientation in relation to each other, each under separate regulatory control.” Ex. 1001, 83:20-25. Silvestre discloses the use of T-VEC. Ex. 1002, 2:33-36 (“oncolytic herpes simplex 1 (T-VEC) was genetically engineered to attenuate the virus virulence, increase selectivity for cancer cells and enhance antitumor immune response (through GM-CSF expression)”); Ex. 1007, ¶229. T-VEC was well known to a POSITA to be an oncolytic HSV with a heterologous GM-CSF gene inserted into the ICP34.5 encoding locus via a complete deletion and replacement of the ICP34.5 gene with a GM-CSF encoding gene. *Id.*; Ex. 1078, p.5763.

In view of this knowledge in the art, it would have been obvious to a POSITA to use the strategy of T-VEC, as disclosed in Silvestre, and to insert the desired heterologous genes into the ICP34.5 locus. Ex.1007, ¶ 230. The simplest and most common method to insert two genes to a locus is in a back-to-back orientation under control of separate regulatory elements. *Id.* Thus, Claim 13’s limitation was common

¹⁴ To be sure, we recognize that Silvestre discloses as background information that an “oncolytic herpes simplex 1 (T-VEC) was genetically engineered to attenuate the virus virulence, increase selectivity for cancer cells and enhance antitumor immune response (through GM-CSF expression).” Ex. 1002, 2:33–36. Petitioner, however, does not rely upon the T-VEC virus to support its anticipation argument based on Silvestre. *See* Pet. 41–52.

knowledge to a POSITA before the earliest priority date. *Id.* A POSITA would have been motivated to use the gene insertion strategy of Claim 13 and would have had reasonable expectation of success in doing so, because insertion into the ICP34.5 locus had already been successfully demonstrated in the T-VEC oncolytic HSV1 virus. *Id.* Moreover, a POSITA would have been motivated to follow this proven insertion strategy because identifying a different gene insertion site would have required a rigorous trial-and-error process. Ex. 1007, ¶¶ 73-81, 87-88, 230-231.

Pet. 54–55.

In this obviousness ground, Petitioner does not cure the deficiencies we have noted for the anticipation ground based on Silvestre discussed above. Further, having considered all the arguments and evidence of record, we also determine that Petitioner has not demonstrated that claim 13 would have been obvious over Silvestre. Moreover, even assuming that Silvestre anticipates or otherwise renders obvious the limitations of claims 1 and 11, we determine that Petitioner has not demonstrated, with sufficient supporting evidence to carry its burden, that this dependent claim would have been obvious with its additional limitations.

Claim 13 requires the two heterologous genes recited in claim 1 (i.e., the GM-CSF and CTLA-4 inhibitor encoding genes) to be inserted into the “ICP34.5” locus of an HSV in a “back-to-back orientation.”¹⁵ Patent Owner

¹⁵ DNA is “read” in a specific direction, just like letters and words in the English language are read from left to right. In particular, during the transcription process (where DNA is converted into mRNA), a gene is read from the 5' end of the DNA molecule towards the direction of its 3' end. See <https://www.ncbi.nlm.nih.gov/Class/MLACourse/Modules/MolBioReview/basepair.html>. As we understand it, the claimed “back-to-back orientation” requires the two genes to be inserted into the same locus with their respective ends oriented in opposite directions as shown by the arrows in Fig. 5B and Fig. 13 of the '513 Patent. Consistent with these figures,

does not dispute Petitioner's asserted motivation to select the ICP34.5 locus as the point of insertion, which is that "insertion into the ICP34.5 locus had already been successfully demonstrated in the T-VEC oncolytic HSV1 virus," and "a POSITA would have been motivated to follow this proven insertion strategy because identifying a different gene insertion site would have required a rigorous trial-and-error process." Pet. 55. We find that a preponderance of the evidence supports Petitioner's contention that a POSITA would have been motivated to start with the ICP34.5 locus given that it was a known locus for insertion in the T-VEC virus. *See* Ex. 1007 ¶ 77 (Dr. Bell asserting that "[t]here are established loci in certain oncolytic viruses that are known to be ideal locations for heterologous gene insertion. For example, inactivation of the neurovirulence factor ICP34.5 in HSV1 has been well-known to direct tumor-specific cell lysis, making HSV1 ICP34.5 knock-outs useful oncolytic virus vectors."); Ex. 1039 (describing "ICP34.5 deleted herpes simplex virus with enhanced oncolytic, immune stimulating, and anti-tumour properties."); Ex. 1106, 97:12–22 (Dr. Chiocca acknowledging that ICP34.5 was a "published insertion locus," and there would be a reasonable expectation of success in inserting heterologous genes into the ICP34.5 site).

But, even with a motivation to select the ICP34.5 locus as the insertion point, Patent Owner contends that Petitioner has not shown that a POSITA would have been motivated to insert the GM-CSF and the CTLA-4 inhibitor encoding genes in a "back-to-back orientation" in relation to each other, which is required by claim 13. PO Resp. 18–23. Patent Owner

however, there is no requirement that the GM-CSF and CTLA-4 inhibitor encoding genes be directly adjacent to each other. This interpretation is not in dispute.

contends that Dr. Bell's testimony that back-to-back insertion is the "simplest" and "most common" method of inserting genes into a locus is uncorroborated, conclusory, and contradicted by his own patent application where he disclosed inserting genes into a locus in the same orientation. *Id.* at 19–20 (citing Ex. 2008, Figs. 38, 60, ¶¶ 276, 2215; Ex. 2009 ¶ 34). Patent Owner further points out that Du, Petitioner's primary reference for its other obviousness ground (discussed below), as well as Petitioner's own BT-001 oncolytic virus, utilized a strategy wherein two genes were inserted into a locus in the same orientation. *Id.* at 20–21 (citing Ex. 1002, 341; Ex. 2009 ¶ 2010; Ex. 2011, Fig. 1I). Patent Owner also points out that Dr. Bell, at his deposition, acknowledged that it was "perhaps an over-statement" that back-to-back insertion was the "simplest and most common" strategy. *Id.* at 22 (citing Ex. 2021, 43:3–5, 10–13, 16–19). Patent Owner argues that Petitioner's mere assertion that the back-to-back insertion strategy was "known" is not enough to show that a POSITA would have been motivated to use it. Sur-Reply 15–16.

Patent Owner additionally argues that Petitioner has not shown that a POSITA would have had a reasonable expectation of success in achieving the claimed invention. PO Resp. 23–27. Patent Owner points out that Petitioner has inconsistently argued, both in the Petition as well as during prosecution of Silvestre, that the claimed technology is difficult and unpredictable. *Id.* (citing Pet. 20, 22, 23; Ex. 1007 ¶¶ 80, 94, 145, 177; Ex. 2015, 8-26-2019 Response at 19). In particular, Patent Owner contends that Petitioner, in the course of arguing lack of enablement for the now disclaimed challenged claims, admitted the difficulty and unpredictability associated with creating a functional oncolytic virus. *Id.* In response to Petitioner's argument that Patent Owner relies upon unclaimed features

(Reply 12–14), Patent Owner contends that it is proper to consider how the overall virus would function as part of the reasonable expectation of success analysis because claim 13 explicitly recites an “oncolytic virus,” which Petitioner has construed in functional terms. Sur-Reply 17–20.¹⁶

a) “Simple” and “Common” Rationale

In our Institution Decision, we took into account Dr. Bell’s testimony that “[w]hen inserting two genes into a given locus, the simplest and most common way to do so is to insert them in a back-to-back orientation under control of separate regulatory elements.” Ex. 1007 ¶ 230; Inst. Dec. 12. Dr. Bell’s testimony was not then rebutted by any expert testimony from Patent Owner or tested through cross-examination at the time. We considered and addressed Patent Owner’s argument that Petitioner improperly relies upon “common sense” to supply a missing limitation. Inst. Dec. 15 (citing *Arendi S.A.R.L. v. Apple Inc.*, 832 F.3d 1355, 1361 (Fed. Cir. 2016)). Based on the record at the time, we determined for purposes of institution that Petitioner provided sufficient evidence beyond merely a conclusory assertion of “common sense” to support its threshold burden for claim 13. *Id.* at 15–16. For instance, we noted that Petitioner cited Exhibit 1084, Fig. 1 as showing back-to-back insertion of two genes at the ICP34.5 locus in an HSV1 virus and invited the parties to further develop the record with regard to that exhibit. *Id.* at 16 n.2.

¹⁶ Patent Owner also argues that objective indicia (specifically, unexpected results) support a finding of non-obviousness. PO Resp. 41–48. Petitioner disagrees. Reply 22–26. We address the parties’ arguments with regard to the unexpected results evidence below, after first addressing the arguments as to motivation and reasonable expectation of success.

Since then, however, Dr. Bell's "simplest and most common" rationale has been significantly undermined by the evidence developed during trial. During his deposition, Dr. Bell backtracked as to exactly how "simple" or "common" this insertion strategy was in the art. The relevant portions of Dr. Bell's deposition transcript are reproduced below:

Q. You don't provide an explanation in your declaration why, in your opinion, the back-to-back orientation is the simplest way of inserting two genes into a given locus, right?

A. *I do not provide an explanation for that, no.*

Q. And so how did you determine that the back-to-back orientation was the most common way to insert two genes into a given locus?

A. Well, it is a simple way, and it is very commonly used. . . .

Q. But . . . it's not the most common way, is that your position?

A. I think in fairness . . . , I would have to say it is very common. . . . There certainly are other instances where people put genes into other vectors in series or antagonistically.

But, in general, when one is going this, it's very commonly done to have two genes back-to-back for convenience so you don't have any interference between two genes. So it's something that's been in the practice in the art for a very long time.

So perhaps an over-statement would say it's the simplest and most common, but, you know, its' certainly simple and common.

Q. What did you say before that?

A. I would say it's a very common way of doing it.

Q. But it's not the most common?

A. I don't think I could say – *maybe I overstated by saying it's the most common because I don't know if anyone has actually made the calculation*, if it has been done.

But if I was a person skilled in the art, it would be one of the most likely ways I would go about doing it. So *perhaps I overstated it by saying it was the most common because I don't think I can quantitate that necessarily*. . . .

Ex. 2021, 41:7–43:19 (emphasis added). Dr. Bell acknowledged that he did not cite any publications to support his opinion regarding back-to-back insertion “because [he] felt it was common knowledge,” and [i]t just didn't occur to [him] to include that citation.” *Id.* at 44:2–21.

In view of the foregoing testimony, Petitioner retreats from its original contention that back-to-back insertion was “[t]he simplest and most common method,” and contends in its Reply that motivation to use that technique comes from that method merely being “simple” and “commonly used.” Reply 9–11. This is a different theory than what was originally presented in the Petition. But even upon considering it, Petitioner has not identified any evidence of record showing exactly how simple or common back-to-back insertion was compared to the other techniques that were known in the art for inserting two genes into the same locus.

Furthermore, Petitioner's motivation for using this technique is contradicted by other evidence of record, including the testimony of Dr. Chiocca regarding examples of other strategies that were used to insert two genes into a given locus of a virus genome. Dr. Chiocca testifies that “there are multiple insertion strategies that one could use and would need to experiment with when inserting two genes in a locus.” Ex. 2009 ¶ 33. For example, according to Dr. Chiocca, “bicistronic insertion” is another common strategy that a POSITA would have employed using “internal ribosomal entry sites” (IRES) or “2A” peptides. *Id.* (citing Ex. 2019).

Dr. Chiocca further notes that Du, the primary reference used in Petitioner's other obviousness grounds, used a 2A sequence between the genes for the heavy and light chains in the oncolytic adenovirus expressing the anti-CTLA-4 antibody to ensure that "[t]he equimolarity of the heavy and light chains is achieved." *Id.* (citing Ex. 1003, 341). Dr. Chiocca's testimony is also supported by other cited evidence of record suggesting that bicistronic insertion (not back-to-back insertion) was in fact the most common way of inserting multiple genes into a locus. Ex. 2019, 2 ("The most common strategy for the expression of multiple genes has been based on internal ribosome entry site (IRES) elements. . . . An alternative to these approaches is the use of ribosomal skipping via 2A peptides.").

Dr. Chiocca also points out that, for Dr. Bell's own oncolytic vaccinia virus construct described in U.S. Patent Application Publication 2022/0056480 A1, Dr. Bell inserted genes into a locus in the same orientation, and thus "did not employ what he claims is the 'simplest' and 'most common' way of inserting two genes into a locus." Ex. 2009 ¶ 34 (citing Ex. 2008,¹⁷ Figs. 38, 60, ¶¶ 275, 2215). This is also true for Petitioner's own BT-001 oncolytic vaccinia virus, which contains genes for the anti-CTLA-4 antibody and GM-CSF inserted into the virus's I4L locus in the same orientation. *Id.* ¶ 35 (citing Ex. 2010, Ex. 2011)¹⁸; *see also id.*

¹⁷ Dr. Chiocca's declaration cites Ex. 2009, but we understand this to be a typographical error.

¹⁸ We are not persuaded by Petitioner's contention that Ex. 2008, Ex. 2010, and Ex. 2011 are entitled to no weight because they were published after the 2016 filing date of the '513 patent. Reply 11. Patent Owner contends that its reliance on post-filing date publications is proper insofar as they are relied upon to contradict Petitioner's assertion that "the simplest and most common method to insert two genes to a locus is in a back-to-back orientation." Sur-Reply 17. We agree. We consider these exhibits, not for

¶ 50 (noting that “Dr. Bell and Petitioners themselves use bicistronic and/or same orientation insertion strategies in their own oncolytic vaccinia viruses”).

We have also considered the deposition testimony of Dr. Chiocca, which Petitioner cites as an admission “that in 2016 back-to-back insertion was a well-known technique.” Reply 10 (citing Ex. 1106, 147:14–148:7). While Dr. Chiocca acknowledged that “it [was] well-known that there was a number of different ways” you could insert heterologous genes into a virus and did not dispute “that back-to-back insertion was a well-known technique,” he also made it clear that back-to-back insertion “was not the most common preferred method” and “there were other common methods as well.” Ex. 1106, 147:14–148:7. Earlier during his deposition, Dr. Chiocca testified on this point as follows:

Q. So if you were looking -- if a person of ordinary skill in the art were looking to insert two heterologous genes, they would have a reasonable expectation of success in using a back-to-back insertion strategy under separate regulatory control?

A. *In a general sense, yes.*

Q. Okay. And then sort of putting that together, by January of 2016, it was known that you could make back-to-

their prior art value, but rather as evidence that call into question Dr. Bell’s credibility as to what techniques may have been considered “simple” and “common” by those skilled in the art. *See Yeda Research v. Mylan Pharm. Inc.*, 906 F.3d 1031, 1041 (Fed. Cir. 2018) (“[N]on-prior art evidence of what was known cannot be applied, independently, as teachings separately combinable with other prior art, but can be relied on for their proper supporting roles, e.g., indicating the level of ordinary skill in the art, what certain terms would mean to one with ordinary skill in the art, and how one with ordinary skill in the art would have understood a prior art disclosure.” (quotation and citation omitted)).

back insertion of multiple heterologous genes into the ICP 34.5 site, correct?

A. *No, not to my knowledge.*

Q. Okay.

A. *I think there was maybe just one paper, actually by the same group back in the mid-2000s that they did that.*

Q. So it had been done by then?

A. There was one paper, right. And actually I only found out in the course of my -- found out about that paper in the course of this.

Q. But as we discussed --

A. *It was not a commonly known strategy, let me put it this way.*

Q. It had been used before?

A. Yeah.

Q. But it might not have been used a lot?

A. *Right.*

Id. at 98:16–99:21 (emphasis added). As such, Dr. Chiocca made it clear that although a POSITA may have had “in a general sense” a reasonable expectation of success in using a back-to-back insertion strategy, “[i]t was not a commonly known strategy.” *Id.* When considered in its totality, Dr. Chiocca’s deposition testimony is consistent with and does not undermine the opinions set forth in his declaration.

We therefore credit the testimony of Dr. Chiocca over the contradictory, unsupported, and admittedly over-stated testimony of Dr. Bell. We note that, despite having the opportunity to do so, Petitioner did not offer any expert declaration with its Reply to rebut Dr. Chiocca’s testimony. Accordingly, Petitioner does not persuade us that a POSITA would have chosen to insert the GM-CSF and CTLA-4 inhibitor genes into

the ICP34.5 locus of an HSV in a back-to-back orientation based on the simplicity or commonality of that technique. Other than its having been known and the now-diminished point that it was the most common and simplest approach, Petitioner does not identify any other benefits to be gained from using back-to-back insertion over other types of insertion strategies that may have made it an attractive choice for the ordinarily skilled artisan at the time of the invention.¹⁹ *See generally* Pet.

b) “Known-Technique” Rationale

Regardless of how “simple” or “common” back-to-back insertion was in the art, motivation to use that technique may come from the mere fact that it was one “known” option among a “finite number of identified, predictable solutions” within a POSITA’s technical grasp. *KSR*, 550 U.S. at 421. To help us assess whether this rationale applies here, we asked the parties to address the impact of two recent decisions on the obviousness analysis in this case: the Federal Circuit’s decision in *Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373, 1380 (Fed. Cir. 2023) (“*Intel*”) and the Director-issued opinion in *Xerox Corp. v. Bytemark, Inc.*, IPR2022-00624, Paper 12 (PTAB Feb. 10, 2023) (“*Xerox*”). Paper 33 (authorizing supplemental briefing).

In *Intel*, the Federal Circuit emphasized that “[t]he motivation-to combine analysis is a flexible one” and that “‘universal’ motivations known in a particular field to improve technology” can be sufficient ‘*even absent any hint of suggestion in the references themselves.*’” *Intel*, 61 F.4th at

¹⁹ At his deposition, Dr. Bell alluded to the avoidance of “interference” between two genes as another reason to use back-to-back insertion. Ex. 2021, 42:19–22, 43:23–44:1. Dr. Bell, however, did not make any reference to avoiding interference in his declaration, nor did Petitioner rely upon this rationale in any of its briefs in this proceeding. Accordingly, we do not consider that as a separate reason to use back-to-back insertion.

1379–80 (citing *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 797–99 (Fed. Cir. 2021)²⁰). The *Intel* court held that it was improper for the Board to reject the so-called “known-technique” rationale in that case, which is based on *KSR*’s statement that “if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill.” *Id.* at 1380 (citing *KSR*, 550 U.S. at 417); *see also id.* (“And if there is a known technique to address a known problem using ‘prior art elements according to their established functions,’ then there is a motivation to combine.”) (citing *Qualcomm*, 21 F.4th at 799–800).

In *Xerox*, the Director on *sua sponte* review, affirmed a Board decision denying institution of an *inter partes* review. *Xerox*, Paper 12. In doing so, the Director emphasized that the rationale set forth in the petitioner’s expert declaration “is an exact restatement of the [p]etition’s arguments without any additional supporting evidence or reasoning.” *Id.* at 4. And the Director rejected the petitioner’s argument in that case that “the claimed account blocking technique for fraud, which was not expressly disclosed in the prior art, is ‘common sense’ based on current known fraud detection technologies” and that “[t]here are only a finite number of predictable, known options for identifying and blocking fraudulent purchasers.” *Id.* The Director further determined that “the Board was correct in giving little weight to [p]etitioner’s expert because the expert declaration merely offered conclusory assertions without underlying factual

²⁰ As we further discuss *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784 (Fed. Cir. 2021) below, we will refer to that case as “*Qualcomm*” to distinguish it from *Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373 (Fed. Cir. 2023).

support.” *Id.* at 5. The same day, the Director designated the underlying panel decision—which reiterated many of the same points as the Director—as precedential. *Xerox*, Paper 9 (Aug. 24, 2022; designated precedential Feb. 10, 2023).

In its supplemental brief, Petitioner contends that *Intel* further supports its contention that a motivation exists to use the known back-to-back insertion strategy to arrive at claim 13. Paper 34, 3–5. Petitioner, however, argues that *Xerox* is inapplicable because: 1) “no dispute exists as to the facts proffered in Dr. Bell’s Declaration” and 2) “Dr. Bell’s declaration includes ‘additional supporting evidence [and] provide[s] ... technical reasoning to support his statement.’” *Id.* at 1–2 (citing *Xerox*, 15). In any event, Petitioner contends that application of *Xerox* does not impact the result in the present case because “the other evidence of record would be sufficient to demonstrate that this technique was well known.” *Id.* at 3.

On the other hand, Patent Owner, in its supplemental brief, contends the opposite. That is, according to Patent Owner, the conclusory nature of Dr. Bell’s assertion as to what is allegedly the “simplest and most common” method for inserting two genes into a locus is similar to what was found insufficient by the Director in *Xerox*. Paper 35, 1. Patent Owner contends that Dr. Bell’s admitted “over-statement” is “particularly problematic in cases where, like here, expert testimony is offered not simply to provide a motivation to combine prior-art teachings, but rather to supply a limitation missing from the prior art.” *Id.* (citing *Xerox*, 16). Patent Owner, however, contends that *Intel*’s “known-technique” rationale does not help Petitioner because the technology in this case is unpredictable and complicated, and, unlike in *Intel*, the asserted prior art does not actually teach the missing element. *Id.* at 3–4.

We find that the evidence relied upon by Petitioner in this case is more akin to what was found insufficient in *Xerox* than what was found sufficient in *Intel* to support an obviousness determination based on the known-technique rationale. In particular, as with the petitioner's expert declaration in *Xerox*, we find that Dr. Bell's declaration in this case merely offers conclusory assertions without underlying factual support as to what techniques were known in the art about insertion of multiple heterologous genes into the same locus. Indeed, Dr. Bell fails to cite any evidence in his declaration, either in paragraph 230 or the cross-referenced paragraphs 73–81, and 87–88, to support his opinion that back-to-back insertion is the “simplest and most common” and “most straightforward” strategy a POSITA would use to insert two genes into a given locus. Ex. 1007 ¶¶ 73–81, 87–88, and 230.²¹

Petitioner, in its Reply, points out that Dr. Bell made a reference during his deposition to an article by Simpson²² that “describes a herpes virus with an insertion into the ICP34.5 locus of a dell fusion protein and a suicide gene that are put in back-to-back when two separate regulatory elements from 2006,” which he “should have probably cited [at paragraph 230] as well.” Reply 10–11; Ex. 2021, 44:2–21. Simpson was only cited in the Petition and Dr. Bell's declaration for its teachings concerning the

²¹ As noted in our Institution Decision, Dr. Bell did cite references teaching the use of the ICP34.5 locus. Inst. Dec. 15; Ex. 1007 ¶ 77 (citing Ex. 1039, 292, Ex. 1040, 4–7, Ex. 1041, 575). But Petitioner and Dr. Bell did not point to any teachings in those cited references about the insertion of multiple genes in a back-to-back orientation into that locus, and we discern no such teachings based on our own review.

²² Simpson *et al.*, *Combination of a Fusogenic Glycoprotein, Prodrug Activation, and Oncolytic Herpes Simplex Virus for Enhanced Local Tumor Control*, 66(9) CANCER RES. 4835-42 (May 1, 2006) (Ex. 1084).

“GALV” protein recited in (now disclaimed) claim 5, and not for the back-to-back limitation of claim 13. Pet. 53, 70 (citing, *inter alia*, Ex. 1084); Ex. 1007 ¶¶ 198, 227. As such, we are not persuaded that Dr. Bell’s offhanded citation to Simpson during his deposition is sufficient to carry Petitioner’s burden of proof for claim 13.²³

Moreover, Simpson refers to back-to-back insertion only once, when it states that the “OncoVEX^{GALV/CD}” viral construct “expresses both genes in a back to back orientation,” wherein “the genes were inserted to replace ICP34.5.” Ex. 1084, 4837. Importantly, Simpson does not indicate *why* the genes were inserted in a back-to-back orientation, if there was a reason, nor does it identify any problem that would be solved by using that method of insertion. Nor does Petitioner identify any other evidence of record suggesting that a particular problem would be solved or addressed or advantage gained by the method of insertion. At the oral hearing, Petitioner’s counsel indicated that the “known problem” solved by back-to-back insertion was “simply inserting genes into a virus.” Tr. 11:3–6; *see*

²³ At his deposition, Dr. Bell was also asked during re-direct by Petitioner’s counsel about certain additional references that allegedly taught back-to-back insertion. Ex. 2021, 57:20–58:1, 58:18–21. Petitioner’s counsel brought up this testimony and the new references during the oral hearing, but we decline to consider this evidence as it was not previously cited by Petitioner. We previously cautioned the parties that oral argument is not an opportunity to raise new arguments and evidence. Paper 30 (Oral Hearing Order), 3; *see also* CTPG, 85 (“[N]o new evidence may be presented at the oral argument.”); *Dell Inc. v. Acceleron, LLC*, 884 F.3d 1364, 1369 (Fed. Cir. 2018) (holding that the Board is obligated under its own regulations to dismiss untimely argument “raised for the first time during oral argument”); *St. Jude Med., Cardiology Div., Inc. v. The Bd. of Regents of the Univ. of Mich.*, IPR2013-00041, Paper 65, 2–3 (PTAB Jan. 27, 2014) (explaining that “new” evidence includes evidence already of record but not previously discussed in any paper of record).

also id. at 11:13–23 (characterizing the “problem” to be solved as “inserting two genes into the genome of a virus” and “how do you get two genes into the virus without disrupting its function”). But this was not a “problem” identified in any prior art reference or in Dr. Bell’s declaration. Petitioner’s counsel also acknowledged that “frankly, I think there’s very little challenge in inserting in two genes,” and “[y]ou can do it a number of ways, all were known techniques, all were suitable techniques and I think that is really what the *Intel* case is about.” *Id.* at 11:21–24. We are not persuaded that merely inserting two genes into a virus constitutes a problem that would have been particularly solved by back-to-back insertion.

By contrast, in *Intel* and *Qualcomm*, the Federal Circuit noted that motivation derived from the fact that the prior art expressly addressed a known problem that was solved using the different known techniques. *See Intel*, 61 F.4th at 1380 (“That Kabemoto and Bauman address the same problem and that Bauman’s cache was a known way to address that problem is precisely the reason that there’s a motivation to combine under *KSR* and our precedent.”); *id.* at 1381 (“It’s enough for Intel to show that there was a known problem of cache coherency in the art, that Bauman’s secondary cache helped address that issue, and that combining the teachings of Kabemoto and Bauman wasn’t beyond the skill of an ordinary artisan.”); *Qualcomm*, 21 F.4th at 799 (noting motivation to use secondary reference’s “prior art switch to solve a known problem—the ‘feedthrough problem’—that was an expected negative consequence of” modifying the primary reference).²⁴

²⁴ In *Qualcomm*, the court recognized that certain “generic” reasons may suffice as a motivation for obviousness, but also reiterated that “[c]onclusory expert testimony does not qualify as substantial evidence” to support such a

Accordingly, in the absence of any identification of a known problem to be solved, we are not persuaded that the mere fact that back-to-back insertion may have been known in the art (albeit evidenced by only a single reference) establishes a motivation to use that technique for the insertion of the GM-CSF and CTLA-4 inhibitor genes into the ICP34.5 locus.

c) Reasonable Expectation of Success

In addition to a lack of motivation, Patent Owner contends that Petitioner has not explained how a POSITA, in view of Silvestre, would have had a reasonable expectation of success in achieving the claimed invention. PO Resp. 23–27. We agree.

Reasonable expectation of success is a separate legal requirement for obviousness. *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016). Petitioner’s entire reasonable expectation of success argument for claim 13 is that “[a] POSITA would have been motivated to use the gene insertion strategy of claim 13 and would have had a reasonable expectation of success in doing so, because insertion into the ICP34.5 locus had already been successfully demonstrated in the T-VEC oncolytic HSV1 virus.” Pet. 55. But as noted above, at least in the Petition, Petitioner did not identify any evidence indicating that two genes were successfully inserted into the ICP34.5 locus in a back-to-back orientation. We do not find that the single example of back-to-back insertion taught by Simpson establishes that a POSITA would have a reasonable expectation of success in using that technique for any other heterologous genes.

motivation. 21 F.4th at 797 (citing *TQ Delta, LLC v. Cisco Sys., Inc.*, 942 F.3d 1352, 1358–59 (Fed. Cir. 2019)). This is entirely consistent with both the Director’s opinion in *Xerox* and our conclusion here.

Moreover, even if a POSITA would have been able to insert the two genes in a back-to-back orientation without difficulty, we are not persuaded that a POSITA would have been able to successfully achieve the claimed oncolytic virus as a whole. In the course of arguing lack of enablement, Petitioner acknowledges that the claimed technology is difficult and there is a high level of unpredictability in the art. *See* Pet. 20 (“The Process of Creating A Virus, Especially An Oncolytic Virus, That Expresses Heterologous Genes is Difficult and Unpredictable”); *id.* at 22 (“Producing an oncolytic virus containing multiple heterologous genes is difficult, time consuming, and uncertain.”). This is supported by Dr. Bell’s uncontradicted testimony. *See, e.g.*, Ex. 1007 ¶ 80 (“When expressed at too high a level, some heterologous proteins can be detrimental to replication of the oncolytic virus, even making the virus effectively replication incompetent.”); *id.* ¶ 94 (discussing reasons why “producing an oncolytic virus containing multiple heterologous genes is a difficult, time consuming, and uncertain task with a path that includes many daunting tasks including . . . [determining] whether the location of the insertion will impact the timing of expression of the heterologous gene” and ensuring that the oncolytic virus “actually infects and replicates in tumor cells, such that the tumor cells are killed, and to ensure that it effectively expresses the heterologous genes in the target cancer cells”); *id.* ¶ 145 (stating that extensive experimentation would be required to determine “whether the heterologous genes that are expressed are providing the desired function within the tumor microenvironment,” and “whether the oncolytic virus is effective in treating any particular type of cancer”); *id.* ¶ 177 (“[A] POSITA at the relevant time would have known that creation of oncolytic viruses with heterologous GM-CSF and anti-CTLA-4 encoding genes was a difficult task involving technologies that are

still difficult and unpredictable.”).²⁵ Dr. Chiocca agrees. Ex. 2009 ¶¶ 38–39. As explained by Dr. Chiocca, this uncertainty establishes that there would not have been a reasonable expectation of success for claim 13. *Id.*

Petitioner does not refute the foregoing testimony, but argues in its Reply that Patent Owner “conducted the wrong analysis by focusing not on whether a POSITA could make the virus of Claim 13 from Silvestre, but instead on how the created virus would function.” Reply 12–13. According to Petitioner, “[u]nclaimed features are irrelevant to the reasonable expectation of success analysis.” *Id.* at 13 (citing *BTG Int’l Ltd. v. Amneal Pharms. LLC*, 923 F.3d 1063, 1075–76 (Fed. Cir. 2019)). We are not persuaded by this argument. Claim 13, by incorporating all the features of claims 1 and 11, expressly requires an “oncolytic virus.” See 35 U.S.C. § 112(d). Petitioner’s own construction for “oncolytic virus,” which we have adopted, requires “a virus that infects and replicates in tumor cells, such that the tumor cells are killed.” Pet. 6. This is distinguishable from the *BTG* case cited by Petitioner, where there was no need to demonstrate reasonable expectation of success related to “survival advantage” since it was not required by the claims. As such, the admitted difficulty in creating a functional oncolytic virus is certainly relevant to the reasonable expectation of success analysis for claim 13.

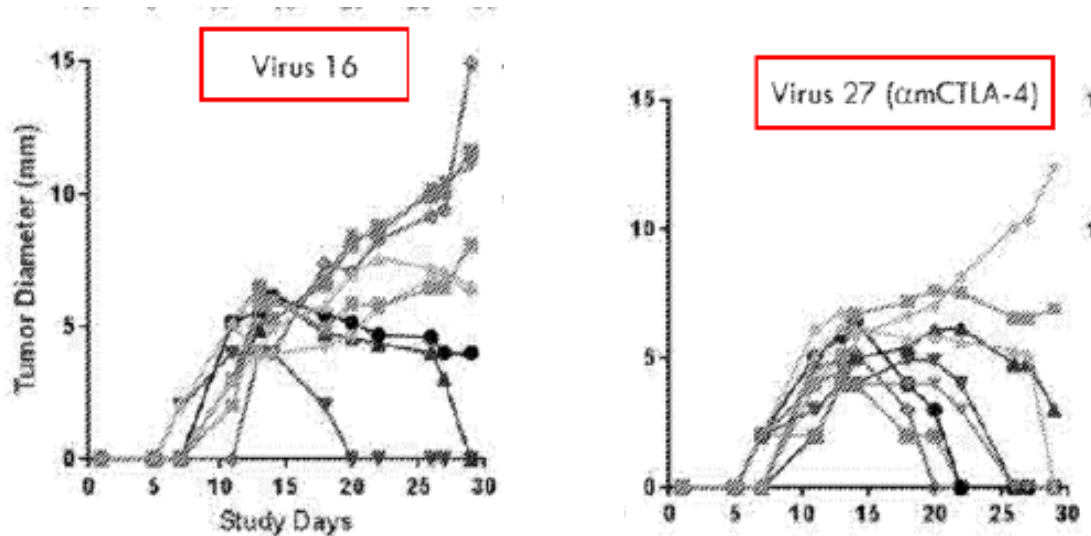
Accordingly, we determine that Petitioner has not demonstrated that a POSITA would have a reasonable expectation of success in making the oncolytic virus of claim 13.

²⁵ In crediting this testimony, we take no position on whether claim 13 is enabled under § 112.

d) Unexpected Results

Although we have already identified deficiencies in Petitioner's obviousness case for claim 13, for the sake of completeness we address the asserted objective indicia of non-obviousness. Patent Owner contends that objective indicia in the form of unexpected results further support a conclusion of non-obviousness. PO Resp. 41–47. In particular, Patent Owner relies upon the experiments described in Example 13 of the '513 patent as showing that “the claimed oncolytic virus demonstrates improved **anti-tumor effects** that are neither taught nor suggested by the prior art.” *Id.* at 41 (citing Ex. 2009 ¶¶ 56–64). Specifically, Example 13 compared the anti-tumor effects of an oncolytic HSV1 expressing an anti-CTLA-4 antibody and GM-CSF (in a back-to-back orientation) as well as GALV (Virus 27) to those of an oncolytic HSV1 expressing GM-CSF and GALV only (Virus 16). Ex. 1001, 30:45–31:14. Mouse “A20” tumors were grown in both flanks of mice, but the oncolytic viruses were injected into only the right flank tumors. *Id.* The effects on tumor growth were observed for 30 days.

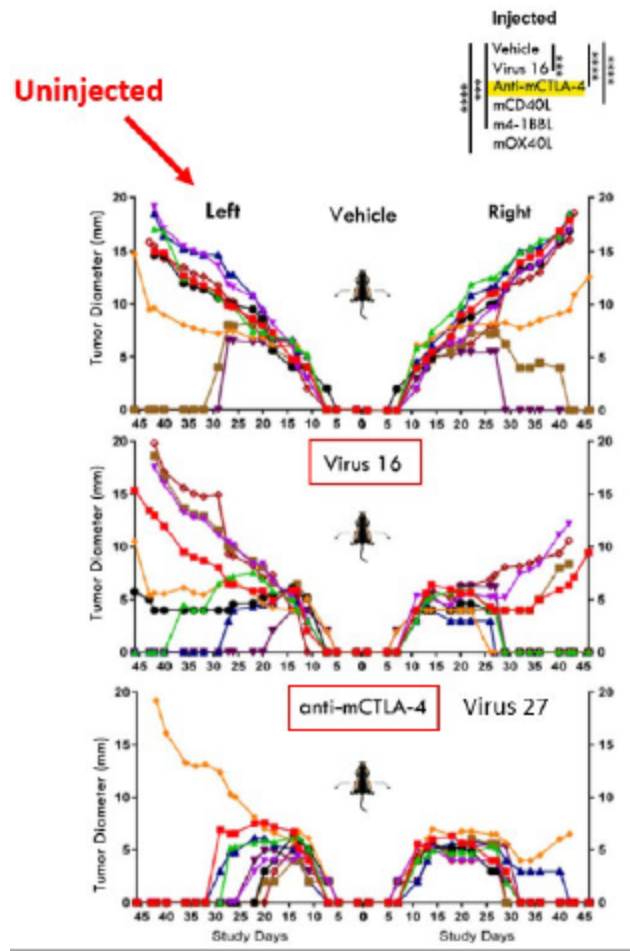
Reproduced below are portions of Figure 16 showing the tumor diameter of the non-injected (left flank) tumors over time following injection of Virus 16 and Virus 27:



Ex. 1001, Fig. 16. The graph on the left side above shows the tumor diameter data for mice injected with Virus 16, while the graph on the right shows the data for mice injected with Virus 27 (each line of the graphs representing a different mouse). As explained by Dr. Chiocca, “while virus 16 (HSV + GM-CSF + GALV) had anti-tumor effects in the non-injected tumor, virus 27 (HSV + GM-CSF + GALV + anti-CTLA-4 antibody) triggered a significantly better response than virus 16.” Ex. 2009 ¶ 58.

Patent Owner also points to similar results from Thomas,²⁶ which includes results from the same experiment disclosed in the '513 patent, but further includes the data for the injected (right flank) tumors and additional data points up to study day 45. PO Resp. 43. An annotated portion of Figure 7(B) from Thomas is reproduced below:

²⁶ Thomas et al., *Development of a new fusion-enhanced oncolytic immunotherapy platform based on herpes simplex virus type 1*, J. FOR IMMUNOTHERAPY OF CANCER 7:214, 1–17 (2019) (Ex. 2013).



Ex. 2013, 15 (Fig. 7(B)). The top-most graph above shows tumor growth on both left and right flanks for uninjected mice, the middle graph shows results for mice injected with Virus 16, and the bottom graph shows results for mice injected with Virus 27. These results are consistent with the data shown in the '513 patent.

Although there was one outlier mouse for which tumor growth went unabated, Dr. Chiocca explains that the data above shows a statistically significant improvement in the antitumor effects of Virus 27. Ex. 2009 ¶¶ 58–59. Specifically, Dr. Chiocca attests that “the expression of anti-CTLA-4 antibody in addition to GM-CSF in Virus 27 significantly increased the antitumor effect, not only in injected tumors (p value of 0.0001 (****)), but also in un-injected tumors (p value of 0.01 (**)) as compared to Virus

16, which showed an antitumor effect only in injected tumors (p value of 0.001 (***)), but not in un-injected tumors,” where P values of less than 0.05 were considered significant. *Id.* Dr. Chiocca further attests that these results would not have been predicted by a POSITA given the complex interplay of the anti-tumor and anti-viral immune responses induced by the oncolytic viruses. *Id.* ¶¶ 60–62. Dr. Chiocca also explained both in his declaration and during his deposition that these results showed that the claimed oncolytic virus exhibited “improved systemic anti-tumor effects,” i.e., an “abscopal” effect on the uninjected tumors. Ex. 2009 ¶¶ 57–58; Ex. 1106, 151:22–152:5 (“The unexpected results to me arises from expression of [anti-]CTLA-4 within the virus giving you a superior . . . abscopal anti-tumor effect -- abscopal means in the contralateral flank.”).

Petitioner does not provide any expert declaration to rebut the foregoing evidence. Rather, Petitioner contends that Du previously taught a “synergistic effect” when oncolytic viruses expressing GM-CSF and anti-CTLA-4 are co-administered, and this was the same “unexpected result” later shown in the ’513 patent. Reply 23 (citing Ex. 1003, 347). Petitioner further contends that, even if there was an unexpected result shown in the ’513 patent, it only relates to the subject matter of claim 1, not claim 13, because the result arises from the administration of an oncolytic virus that expresses both GM-CSF and anti-CTLA-4 as compared to one that does not express anti-CTLA-4. *Id.* at 24. Petitioner cites to deposition testimony of Dr. Chiocca where he allegedly admitted these points. *Id.* at 24–25 (citing Ex. 1106, 151:19–153:9, 156:5–158:6, 158:21–159:13).

We are not persuaded by Petitioner’s attorney arguments against Patent Owner’s unexpected results evidence. In particular, we do not agree

that the unexpected results shown in the '513 patent and Thomas merely reflect what was already shown in Du. Du states the following:

We have examined the synergistic effects of GM-CSF-mediated antitumor activity following the oncolysis of an adenovirus and CTLA4 blockade following infection of SKL001 and SKL002, and resulted in significantly higher antitumor efficacy in the group treated with GM-CSF expressing oncolytic adenovirus (SKL001) and CTLA4 blockade-expressing adenovirus (SKL002) when compared with the group treated with either agent alone, suggesting synergistic antitumor activity of the combination.

Ex. 1003, 347.

While Dr. Chiocca responded “[c]orrect” when asked if Du taught the same “enhanced effect” that was shown in the '513 patent (Ex. 1106, 159:1–13), he also qualified that there were “some serious differences” between the two studies, including the fact that the results shown in Du were based on co-administering two different adenoviruses (as opposed to a single HSV expressing both genes), and that Du did not demonstrate any abscopal effects on uninjected tumors. *See id.* at 157:14–158:2 (answering “I’m trying to tell you that there’s some serious differences” when asked to compare the results of Du with the '513 patent); *id.* at 159:17–160:9 (acknowledging that Du utilized two adenoviruses, each separately expressing GM-CSF and anti-CTLA-4); *id.* at 178:3–10 (“Again, when you look at that experiment [in Du], it is -- pretty much what it shows is an injection into tumor of two adenoviruses . . . They’re not looking at [uninjected tumors],²⁷ they’re not looking at any abscopal effects. So they’re just looking at the direct effect in the tumor.”). We credit this

²⁷ Dr. Chiocca issued an errata replacing “lymphatic cancer” in the original transcript with “uninjected tumors.” Ex. 1106, 235.

undisputed testimony of Dr. Chiocca, and find that the results shown in the '513 patent would not have been expected based on Du's teachings.

We are also unpersuaded by Petitioner's argument that the unexpected results do not have a nexus to claim 13 because they only relate to the recitations of claim 1. Reply 25–26. Patent Owner has not argued that nexus should be presumed in this case. But even without a presumption, the evidence of record supports a conclusion that the unexpected results are the “direct result of the unique characteristics of the claimed invention.” *Fox Factory*, 944 F.3d at 1373–74. It is undisputed that Virus 27, as taught in the '513 patent, embodies the oncolytic virus of claim 13. *See* Ex. 1001, 6:9–15, 7:9–17, Fig. 5B, Fig. 13 (depicting Virus 27 with GM-CSF and anti-CTLA-4 genes inserted into the ICP34.5 locus of HSV1 in a back-to-back orientation). Although Virus 27 includes additional “unclaimed features,” such as the GALV gene, it may still be deemed “coextensive” with the claimed oncolytic virus where such unclaimed features are “insignificant” or otherwise not “critical” to the demonstrated anti-tumor effects. *See Teva Pharms. Int'l GmbH v. Eli Lilly & Co.*, 8 F.4th 1349, 1361 (Fed. Cir. 2021) (explaining that “like the coextensiveness requirement itself, the concept of unclaimed features is best viewed as part of a spectrum”). The test for coextensiveness is not whether the unclaimed feature “materially affects” the functioning of the product. *Id.* Here, Dr. Chiocca has testified that “[a]lthough GALV is expressed by the viruses, its presence plays no role in the antitumor effects of the viruses as GALV causes cell to cell fusion in human cells but not in mouse cells.” Ex. 2009 ¶ 58 n.4 (citing Ex. 1001, 30:15–16, Ex. 1054, 1052–53). Petitioner does not contest this assertion with any evidence. *See WBIP, LLC v. Kohler Co.*, 829 F.3d 1317, 1329

(Fed. Cir. 2016) (“[A] patent challenger cannot successfully rebut the presumption [of nexus] with argument alone—it must present evidence.”).

Petitioner contends that the unexpected results evidence “must have a nexus to the claim limitations, which cannot be satisfied where the secondary consideration ‘actually results from something other than what is both claimed and *novel* in the claim.’” Reply 25 (citing *Kennametal*, 780 F.3d at 1385 (internal quotations omitted)). In *Kennametal*, the patent owner attempted to rely upon unexpected results associated with a combination that was taught by the prior art. 780 F.3d at 1385. Unlike in *Kennametal*, however, we have not determined that claims 1 and 11 are anticipated, such that the requirement for both GM-CSF and anti-CTLA-4 genes in a single HSV would not be considered “novel.” As such, there is no requirement to show that the unexpected results associated with Virus 27 be attributable to the back-to-back feature in particular, as opposed to the oncolytic virus as a whole. *See WBIP*, 829 F.3d at 1331–32 (“Requiring patentees to prove that objective evidence is tied to a specific claim element—and only that claim element—runs counter to the statutory instruction that the obviousness analysis involves determining whether ‘the claimed invention as a whole would have been obvious.’”) (citation omitted).

Accordingly, we determine that the unexpected results evidence presented by Patent Owner sufficiently relates to claim 13 and further supports a conclusion of non-obviousness.

e) Conclusion: Obviousness Based on Silvestre

For the reasons set forth above, we determine that Petitioner has not established that a POSITA would have been motivated, with a reasonable expectation of success, to insert the GM-CSF and anti-CTLA-4 encoding

genes into the ICP34.5 locus of an HSV in a back-to-back orientation to arrive at the oncolytic virus of claim 13 based on Silvestre's teachings. We further determine that Patent Owner has shown unexpected results attributable to the virus of claim 13 that supports the non-obviousness of the claims. Taking all this evidence into consideration, we conclude that Petitioner has not demonstrated by a preponderance of the evidence that claim 13 would have been obvious in view of Silvestre.

F. Obviousness over Du, Choi, and Zitvogel

In Ground 6 of its Petition, Petitioner asserts that claim 13 would have been obvious over the combination of Du, Choi, and Zitvogel. Pet. 74 (Ground 6). Petitioner's arguments also refer to its contentions for disclaimed claims 1 and 11 based on the same prior art combination. *See id.* at 64–68, 74. We therefore consider the parties' arguments concerning the limitations of 1 and 11 before turning to the specific arguments for claim 13.

Petitioner contends that the combination of Du, Choi, and Zitvogel teaches all of the features recited in claims 1 and 11, from which claim 13 depends. Pet. 58–59, 64–66, 74. As to claim 1, Petitioner contends that Du discloses co-administering two oncolytic viruses: a first virus including a heterologous GM-CSF encoding gene and a second virus including a heterologous CTLA-4 inhibitor encoding gene. Pet. 58–59 (citing Ex. 1003, 340). Petitioner argues that a POSITA would have been motivated to “modify Du's teachings to create a single virus with both claimed heterologous genes” by Du alone, and in combination with Choi and Zitvogel. *Id.* at 66.

First, Petitioner argues that “Du provides express motivation” to create a single virus by teaching that co-administering the two viruses provides a synergistic antitumor effect. *Id.* at 59–60 (citing Ex. 1006, 340,

346; Ex. 1007 ¶¶ 247–250). Petitioner contends that a POSITA would have had a reasonable expectation of success in creating a single virus with both heterologous genes inserted, “because it was well known in the art that administering a single oncolytic virus with multiple inserted genes had been done, was preferred, and provided distinct advantages over administering multiple viruses each with different inserted genes, including improved ease of administration.” *Id.* at 61 (citing Ex. 1007 ¶¶ 251–252). Dr. Bell declares that “administering a single virus containing the two heterologous genes would ensure that an infected host cell would receive copies of both heterologous genes and thus should be capable of expressing both proteins,” as opposed to two separate viruses infecting different host cells. Ex. 1007 ¶ 251. Further, Dr. Bell declares that “it was well known in the art that more than one heterologous gene can be inserted into the genome of certain oncolytic viruses (such as HSV1) based on the known genomic payload capacity of those viruses.” *Id.* ¶ 252.

Second, Petitioner argues that a POSITA would have modified Du to create a single virus in view of Choi and Zitvogel. *See* Pet. 66–67. Specifically, Petitioner argues a POSITA would have been motivated by Choi’s teaching of inserting genes encoding GM-CSF and B7-1 into a single oncolytic adenovirus, and Zitvogel’s teaching of co-administering an oncolytic virus that possesses heterologous GM-CSF and an anti-CTLA-4 antibody which can be delivered by viral vector. *Id.* (citing Ex. 1007 ¶ 279; Ex. 1005; Ex. 1004, 3:30–55; 16:11–12; 17:4–5; 18:12–13; 30:10–13; 45:14–46:11). Petitioner further argues that Zitvogel teaches the HSV limitation of claim 11. *See id.* at 74 (citing Ex. 1004, 7:49–52).

With regard to the additional requirements of claim 13, Petitioner argues that Zitvogel discloses the use of T-VEC as an HSV mutant with

defects in the ICP34.5 gene. *Id.* at 74 (citing Ex. 1004, 8:27–34). As with the Silvestre-based obviousness ground, Petitioner also contends that T-VEC was a known oncolytic HSV with a heterologous GM-CSF gene inserted into the ICP34.5 encoding locus via a complete deletion. *Id.* (citing Ex. 1007 ¶ 319; Ex. 1078, 5763). Petitioner further contends (once again) that claim 13’s additional “back-to-back orientation” limitation would have been “[t]he simplest and most common way” of inserting the genes, and that a POSITA would be motivated with a reasonable expectation of success to implement that feature “because insertion . . . into the ICP34.5 locus had already been successfully demonstrated in the T-VEC oncolytic HSV1 virus.” *Id.* (citing Ex. 1007 ¶¶ 319, 73–81, 87–88).

For this obviousness ground, Patent Owner contends that Du teaches the co-administration of two oncolytic viruses (each separately expressing GM-CSF and anti-CTLA-4 antibody), and argues that neither Zitvogel nor Choi would have motivated a POSITA to modify Du’s teachings to arrive at the claimed oncolytic virus. PO Resp. 27–37. In particular, Patent Owner contends that Zitvogel teaches away from making an oncolytic virus encoding an anti-CTLA-4 antibody. *Id.* at 28–31. Patent Owner further contends that Zitvogel has no data on the co-administration of an anti-CTLA-4 antibody and an oncolytic virus encoding GM-CSF, and that the viral-based expression vectors of Zitvogel are not oncolytic viruses. *Id.* at 31–35. Patent Owner also contends that Choi would not have provided the requisite motivation because, although Choi was published over a decade prior to the filing date of the ’513 patent, no POSITA was motivated to make a single oncolytic virus comprising a GM-CSF-encoding gene and a CTLA-4 inhibitor-encoding gene. *Id.* at 35–36. Patent Owner notes that Du chose

to make two separate oncolytic adenoviruses despite being aware of both Choi's work and T-VEC. *Id.* at 36–37.

As to the specific recitations of claim 13, Patent Owner contends that Petitioner has not shown that the “back-to-back” insertion strategy was the “simplest” or “most common.” *Id.* at 37–38. Patent Owner also contends that a POSITA would not have a reasonable expectation of success to arrive at the oncolytic virus of claim 13 because: a) Zitvogel does not disclose any therapeutic results from the coadministration of anti-CTLA-4 antibody and an oncolytic virus that expresses GM-CSF (and instead teaches that sequential administration is superior), b) the results of Du would not have provided a POSITA with a reasonable expectation that administration of a single oncolytic virus, let alone an oncolytic HSV, would lead to the same result, and c) Choi's oncolytic adenovirus contains the two genes inserted in the same orientation, and thus provided no guidance on how to make an oncolytic virus with two genes inserted in a back-to-back orientation. *Id.* at 38–40. Patent Owner also relies upon the same unexpected results evidence discussed above for both sets of obviousness grounds. *Id.* at 41–48.

Having considered all the evidence and arguments of record, we determine that Petitioner has not demonstrated a sufficient motivation, with a reasonable expectation of success, to arrive at the oncolytic virus of claim 13 based on the combination of Du, Choi, and Zitvogel.

The parties' arguments concerning the “back-to-back” limitation of claim 13 based on this combination are similar to what we have already considered above with regard to the Silvestre-based obviousness ground. Petitioner does not identify any additional teachings in Du, Choi, or Zitvogel concerning this limitation, but rather once again asserts that the technique would have been obvious based on it being the “simplest and most common”

way of inserting multiple genes and that the ICP34.5 locus was a known locus for insertion. Pet. 74. Accordingly, we find that Petitioner has not demonstrated that a POSITA would have been motivated to utilize the back-to-back insertion strategy based on either the “simple” and “common” rationale or the “known technique” rationale discussed above. Additionally, for the reasons explained above, we determine that Petitioner has not demonstrated a reasonable expectation of success in view of the admitted difficulty in creating the functional oncolytic virus required by claim 13. We further determine that the unexpected results that we have considered above also support a conclusion of non-obviousness for this ground. We do not repeat those findings and conclusions here. We address, however, certain additional issues specific to the references relied upon for this combination.

Petitioner recognizes that Du teaches synergistic effects only based on the administration of two separate adenoviruses each separately expressing the GM-CSF and anti-CTLA-4 genes. Pet. 60–61; Ex. 1003, 346–47. We are not persuaded that a POSITA would have been motivated to modify Du’s teachings to create a single virus with both claimed heterologous genes. Contrary to Petitioner’s argument, we find nothing in the cited references that would suggest the same synergistic effect could be achieved with a single virus, let alone specifically a single HSV with both genes inserted in a back-to-back orientation in the ICP34.5 locus. Dr. Bell does not cite any evidence to support his assertion that “[a] POSITA would have further been motivated to express both genes in a single virus because it was well known that administration of a single virus is easier than co-administering two separate viruses.” Ex. 1007 ¶ 251. Accordingly, we find

his testimony conclusory and give it little weight. *See Xerox*, Paper 12 at 4–5.

Petitioner relies upon Choi for its teaching of a single oncolytic virus that expresses two heterologous genes. But Choi only discusses the benefits of using an oncolytic adenovirus expressing both GM-CSF and B7-1, which are both proteins exhibiting antitumor effects. Ex. 1005, 1010 (explaining that GM-CSF “is one of the most potent stimulators of a specific and long-lasting antitumor immunity” and “the B7 family has also been shown to play an integral role in mediating an antitumor response”). Choi does not suggest that the same (or improved) antitumor effects could be obtained by replacing the B7-1 gene with a different gene encoding an immune checkpoint modulator such as an anti-CTLA-4 antibody. While Dr. Bell points out that it was well-known in the art that B7-1 also binds to CTLA-4, and thus its gene is a “possible inhibitor-encoding gene” (Ex. 1007 ¶ 272; Ex. 1085, 401), Petitioner has not asserted—and the evidence of record does not otherwise suggest—that Choi’s B7-1 gene would satisfy the claimed CTLA-4 inhibitor gene. Indeed, B7-1 is not described as an immune checkpoint modulator, or as otherwise equivalent to an anti-CTLA-4 antibody, in any of the other references we have considered. Furthermore, Choi’s oncolytic virus contains the two genes inserted in the same orientation, not in a back-to-back orientation. Ex. 1005, Fig. 1; Ex. 2009 ¶ 53. We are persuaded by Patent Owner’s contention that lack of motivation is also evidenced by the fact that Du was clearly aware of Choi’s work. *See* Ex. 2009 ¶ 49; Ex. 1002, References 15 and 16. Although Choi predated Du by about eight years, Du nonetheless chose to use two separate oncolytic viruses rather than a single virus expressing both genes.

Zitvogel teaches a combination therapy based on administering (i) an immune checkpoint modulator, such as an anti-CTLA-4 antibody, which can be delivered via insertion into a viral vector, and (ii) an oncolytic virus that possesses a therapeutic gene, such heterologous GM-CSF. Ex. 1004, 12:9–22:43. Although Zitvogel indicates that sequential administration is preferred (*id.* at 26:51–56), it also teaches that “[t]he oncolytic virus and the one or more immune checkpoint modulator(s) may be administered together in a single composition or concurrently in separate compositions.” *Id.* at 22:44–47. As such, we are not necessarily persuaded by Patent Owner’s contention that Zitvogel teaches away from a single oncolytic virus that encodes both GM-CSF and an anti-CTLA-4 antibody. *See Syntex (U.S.A.) LLC v. Apotex, Inc.*, 407 F.3d 1371, 1380 (Fed. Cir. 2005) (“A statement that a particular combination is not a preferred embodiment does not teach away absent clear discouragement of that combination.”); *In re Fulton*, 391 F.3d 1195, 1201 (Fed. Cir. 2004) (“[P]rior art’s mere disclosure of more than one alternative does not constitute a teaching away from any of these alternatives because such disclosure does not criticize, discredit, or otherwise discourage the solution claimed.”).

Nonetheless, we take into consideration that Zitvogel’s data did not in fact show a synergistic effect or therapeutic benefit from the combination of an anti-CTLA-4 antibody and an oncolytic virus that encodes GM-CSF, as acknowledged by Dr. Bell. Ex. 2021, at 23:12-17 (responding “[t]he figures as presented in Zitvogel do not have that scenario” when asked if the examples in Zitvogel show synergy between anti-CTLA-4 antibody and an oncolytic virus that encodes GM-CSF). Dr. Chiocca attests without dispute that this data only relates to “WRTG17137,” which is an oncolytic vaccinia virus that encodes FCU-1, not an HSV that encodes GM-CSF. Ex. 2009

¶ 45; Ex. 1004, 30:31–41. As further explained by Dr. Chiocca, the viral vectors taught by Zitvogel for expression of the anti-CTLA-4 antibody are not oncolytic viruses because they are replication deficient and do not kill the cells. Ex. 2009 ¶¶ 46–48. Again, this testimony has not been rebutted by Petitioner’s expert. We therefore find that Zitvogel would not have motivated a POSITA to modify Du to include both genes in a single oncolytic HSV.

Taking all this evidence into consideration, we conclude that Petitioner has not demonstrated by a preponderance of the evidence that claim 13 would have been obvious in view of Du, Choi, and Zitvogel.

III. CONCLUSION

After reviewing the arguments and evidence of record, we determine that Petitioner has not demonstrated by a preponderance of the evidence that claim 13 is rendered obvious based on Silvestre, or based on the combination of Du, Choi, and Zitvogel.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
13	103	Silvestre		13
13	103	Du, Choi, Zitvogel		13
Overall Outcome				13

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not demonstrated by a preponderance of the evidence that claim 13 of the '513 patent is unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, any party to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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FOR PETITIONER:

Erin Dunston
Travis Bliss
Aaron Pereira
Stephany Small
PANITCH SCWARZE BELISARIO & NADEL LLP
edunston@panitchlaw.com
tbliss@panitchlaw.com
ssmall@panitchlaw.com
apereira@panitchlaw.com

FOR PATENT OWNER:

Dion Bregman
Alexander Stein
Christopher Betti
Kelly Plummer
Maria Doukas
Guylaine Hache
MORGAN, LEWIS & BOCKIUS LLP
dion.bregman@morganlewis.com
alexander.stein@morganlewis.com
christopher.betti@morganlewis.com
kelly.plummer@morganlewis.com
maria.doukas@morganlewis.com
guylaine.hache@morganlewis.com